

Introduction to Statistics

Agenda

- ❑ What is Statistics?
- ❑ Data Types & Measurement Level
- ❑ Types of Statistics
 - Descriptive Vs. Inferential
- ❑ Descriptive Statistics
 - Types of Descriptive Statistics
 - Measures of Central Tendency
 - Measure of Dispersion
- ❑ Population VS. Sampling
- ❑ Inferential Statistics
 - Data Distribution
 - Hypothesis Testing
- ❑ Regression
 - Types of Regression

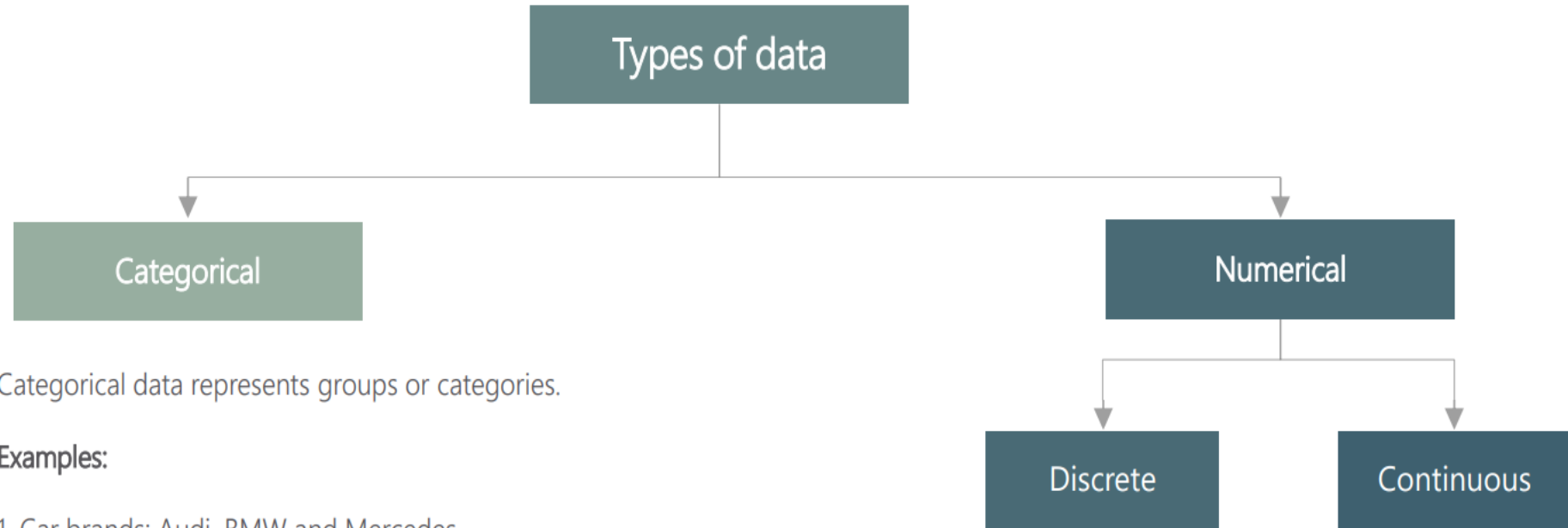
What is Statistics?

The science of collecting, organizing, presenting, analyzing, and interpreting data to assist in making more effective decisions.



Data Types & Measurement Level

Data Types



Categorical data represents groups or categories.

Examples:

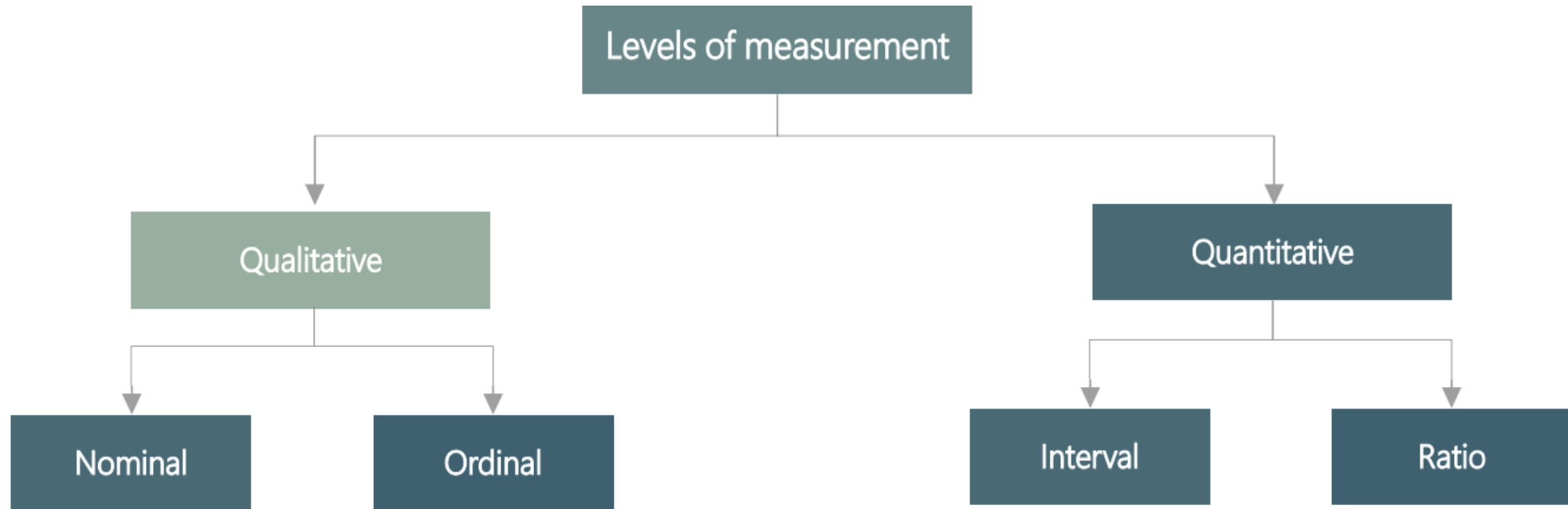
1. Car brands: Audi, BMW and Mercedes.
2. Answers to yes/no questions: yes and no

Numerical data represents numbers. It is divided into two groups: discrete and continuous. Discrete data can be usually counted in a finite matter, while continuous is infinite and impossible to count.

Examples:

Discrete: # children you want to have, SAT score
Continuous: weight, height

Levels of Measurement



There are two qualitative levels: nominal and ordinal. The nominal level represents categories that cannot be put in any order, while ordinal represents categories that **can** be ordered.

Examples:

Nominal: four seasons (winter, spring, summer, autumn)

Ordinal: rating your meal (disgusting, unappetizing, neutral, tasty, and delicious)

There are two quantitative levels: interval and ratio. They both represent "numbers", however, ratios **have a true zero**, while intervals don't.

Examples:

Interval: degrees Celsius and Fahrenheit

Ratio: degrees Kelvin, length

Quantitative Variables

Quantitative variables are characteristics that can be expressed in **numbers**. For example, weight, height, and length can all be written numerically.

Name	Weight	Height	Length
Kiara	4.5 lb	25 cm	12 cm
Simba	5 lb	27 cm	15 cm
Rocky	4.8 lb	35 cm	16 cm
Nala	4.6 lb	31 cm	14 cm
Bruno	5 lb	33 cm	13 cm

Quantitative Variables

- **Continuous Variables:**
 - Contain measurements with decimal precision.
 - **Examples:** Height of individuals, weight of a bag of apples, time taken to run a marathon.
 - **Characteristics:** Measurable, infinitely many possible values, can include fractions/decimals.
- **Discrete Variables:**
 - Contain counts that must be whole integer values.
 - **Examples:** The number of members in a person's family, or the number of goals a basketball team scored in a game.
 - **Characteristics:** Countable, often integers, clear gaps between values.

Qualitative Variables

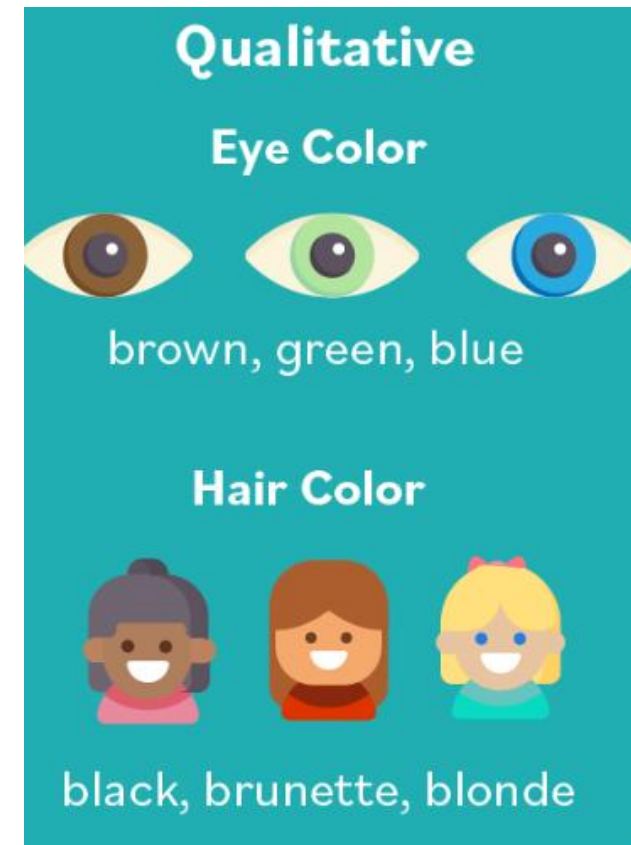
Qualitative variables are characteristics of an individual or object which can only be expressed in **words**. Some examples include ethnicity, profession, or gender.

Ordinal Variables:

- Variables that are groups containing an inherent ranking.
- **Examples:** Education level (high school, bachelor's, master's), customer satisfaction (dissatisfied, neutral, satisfied).

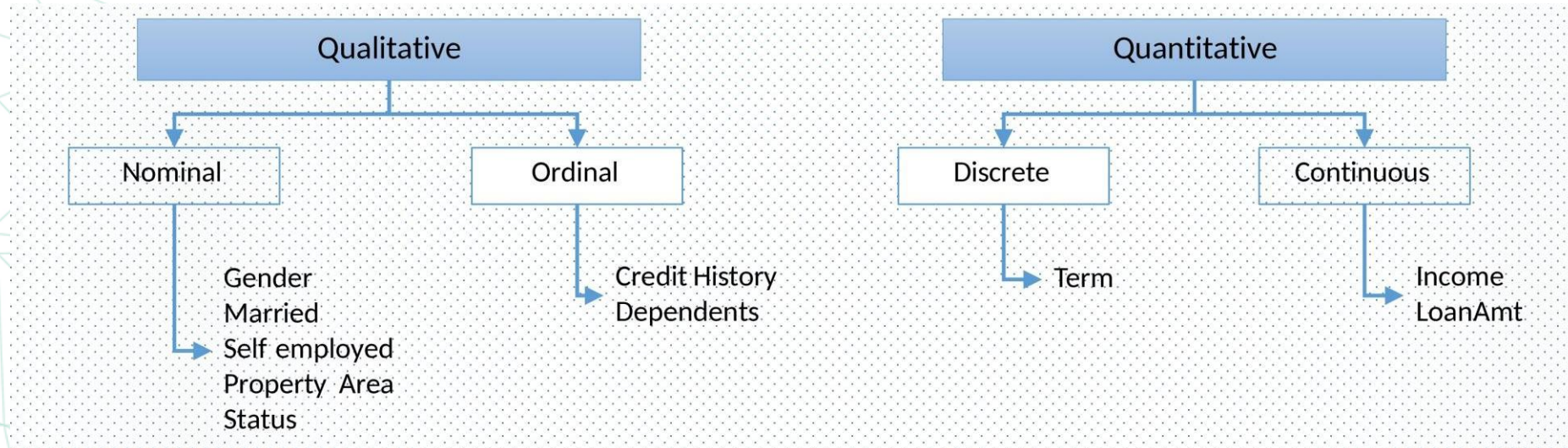
Nominal Variables:

- Variables made up of categories without an inherent order.
- **Examples:** Gender (male, female), eye color (blue, green, brown).



Understanding the Variables Using Dataset

Loan_ID	Gender	Married	Dependents	Self_Employed	Income	LoanAmt	Term	CreditHistory	Property_Area	Status
LP001002	Male	No	0	No	\$5,849.00		60	1	Urban	Y
LP001003	Male	Yes	1	No	\$4,583.00	\$128.00	120	1	Rural	N
LP001005	Male	Yes	0	Yes	\$3,000.00	\$66.00	60	1	Urban	Y
LP001006	Male	Yes	2	No	\$2,583.00	\$120.00	60	1	Urban	Y



Understanding the Variables Using Dataset

Loan_ID	Gender	Married	Dependents	Self_Employed	Income	LoanAmt	Term	CreditHistory	Property_Area	Status
LP001002	Male	No	0	No	\$5,849.00		60	1	Urban	Y
LP001003	Male	Yes	1	No	\$4,583.00	\$128.00	120	1	Rural	N
LP001005	Male	Yes	0	Yes	\$3,000.00	\$66.00	60	1	Urban	Y
LP001006	Male	Yes	2	No	\$2,583.00	\$120.00	60	1	Urban	Y

• Predictor/Independent

- Gender
- Married
- Dependents
- Self_Employed
- Income
- LoanAmt
- Term
- CreditHistory
- PropertyArea

• Target/Dependent

- Status

Data Type

• Character/String

- Gender
- Married
- Self_Employed
- Property_Area
- Status

• Numeric

- Dependents
- Income
- LoanAmt
- Term
- CreditHistory

Types of Statistics

Descriptive Vs. Inferential

Types of Statistics

- **Descriptive Statistics**

The methods of organizing, summarizing, and presenting data in an informative way.
Organizing and summarizing data with frequency tables and frequency distributions.
Presenting frequency tables and distributions with charts and graphs.
Measures to summarize the characteristics of data.

- **Inferential Statistics**

The methods used to estimate population parameters on the basis of a sample statistic. To make inferences (statements) about a population based on a sample, the following concepts are used:

- Probabilities and Probability Distribution

- Sampling

- Estimation

- Hypothesis Testing

- Correlation and Regression



Descriptive Statistics

Types of Descriptive Statistics

Types of Descriptive Analysis



Types of Descriptive Statistics

- **Measures of Frequency:**
 - Describe how often certain values or ranges of values occur within a dataset.
 - They are used to understand the distribution and occurrence patterns of the data.
- **Measures of Central Tendency:**
 - These are ways of describing the central position of a frequency distribution for a group of data, such as **mean**, **median**, and **mode**.
- **Measures of Dispersion or Variation:**
 - These are ways of summarizing a group of data by describing how spread out the scores are.
 - Examples include **range**, **interquartile range**, **variance**, **percentile**, **quartile**, and **standard deviation**.

Frequency Table

- A **frequency table** lists a set of values and how often each one appears.
- **Frequency** is the number of times a specific data value occurs in your dataset.
- These tables help you understand which data values are **common** and which are **rare**.
- They organize your data and are an effective way to present the results to others.
- **Frequency tables** are also known as **frequency distributions** because they allow you to understand the **distribution** of values in your dataset.

Type of Pet	Frequency
Dog	18
Cat	12
Fish	1
None	10

Heights M	Frequency
1.34 - 1.39	4
1.40 - 1.45	14
1.46 - 1.51	31
1.52 - 1.57	19
1.58 - 1.63	14
1.64 - 1.69	6

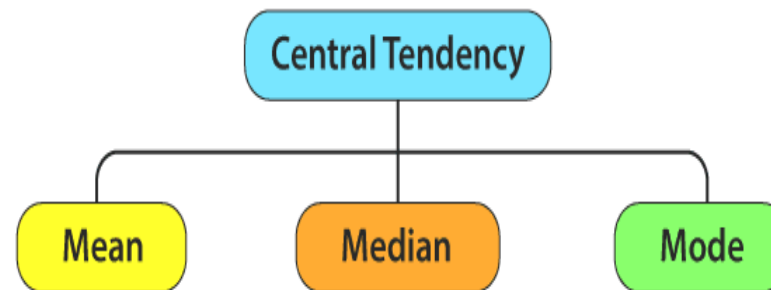
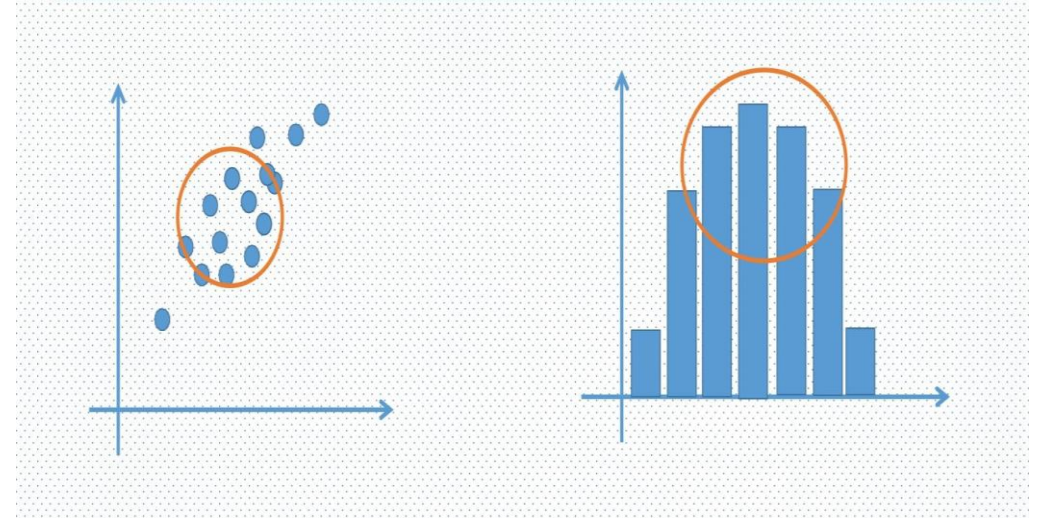
Satisfaction	Frequency
Very satisfied	59
Satisfied	42
Neutral	12
Dissatisfied	8
Very dissatisfied	5

Descriptive Statistics

Measures of Central Tendency

Central Tendency of Data

- ❑ A measure of **central tendency** is a single value that attempts to describe a set of data by identifying the **central position** within that set of data.
- ❑ Measures of Central Tendency:



Mean

- **Mean** is the sum of all the values in the dataset divided by the number of values in the dataset. It is also called the **Arithmetic Average**. **Mean** is denoted as $\bar{x}$ and is read as **x bar**.

$$\text{Mean } (\bar{x}) = \frac{\text{Sum of Values}}{\text{Number of Values}}$$

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N}$$

Mean

The mean is the most widely spread measure of central tendency. It is the simple average of the dataset. **Note: easily affected by outliers**

- **Importance:** The mean provides a central value for a dataset, useful for comparing different data sets and understanding the general trend.
- **Application:** Used in almost all fields such as finance, economics, healthcare, and social sciences to summarize data.

Example Data 1 [-10, 0, 10, 20, 30]

For data 1: $(-10 + 0 + 10 + 20 + 30) / 5 = \underline{10}$

Example Data 2 [8, 9, 10, 11, 12]

For data 2: $(8 + 9 + 10 + 11 + 12) / 5 = \underline{10}$

Median

- **Median** is the middle value for sorted data. The sorting of the data can be done either in **ascending order** or **descending order**. A **median** divides the data into two equal halves.
- In an ordered dataset, the median is the number at position $\frac{n+1}{2}$. If this position is not a whole number, the median is the simple average of the two numbers at positions closest to the calculated value.

Example

Facebook is a popular social networking website. Users can add friends and send them messages, and update their personal profiles to notify friends about themselves and their activities. A sample of 10 adults revealed they spent the following number of hours last month using Facebook.

3	5	7	5	9	1	3	9	17	10
---	---	---	---	---	---	---	---	----	----

Find the median number of hours.

Solution

Note that the number of adults sampled is even (10). The first step, as before, is to order the hours using Facebook from low to high. Then identify the two middle times. The arithmetic mean of the two middle observations gives us the median hours. Arranging the values from low to high:

1	3	3	5	5	7	9	9	10	17
---	---	---	---	---	---	---	---	----	----

The median is found by averaging the two middle values. The middle values are 5 hours and 7 hours, and the mean of these two values is 6. We conclude that the typical Facebook user spends 6 hours per month at the website. Notice that the median is not one of the values. Also, half of the times are below the median and half are above it.

Median

The median is the midpoint of the ordered dataset. While It is not as popular as the mean, it is often used in academia and data science because it is not affected by outliers

- **Importance:** The median gives the middle value of a dataset, making it useful for understanding the distribution of data, especially when the data is skewed.

- **Application:** Often used in income data, housing prices, and other scenarios where outliers may skew the mean.

1, 3, 3, **6**, 7, 8, 9

Median = **6**

Example :

1, 2, 3, **4**, **5**, 6, 8, 9

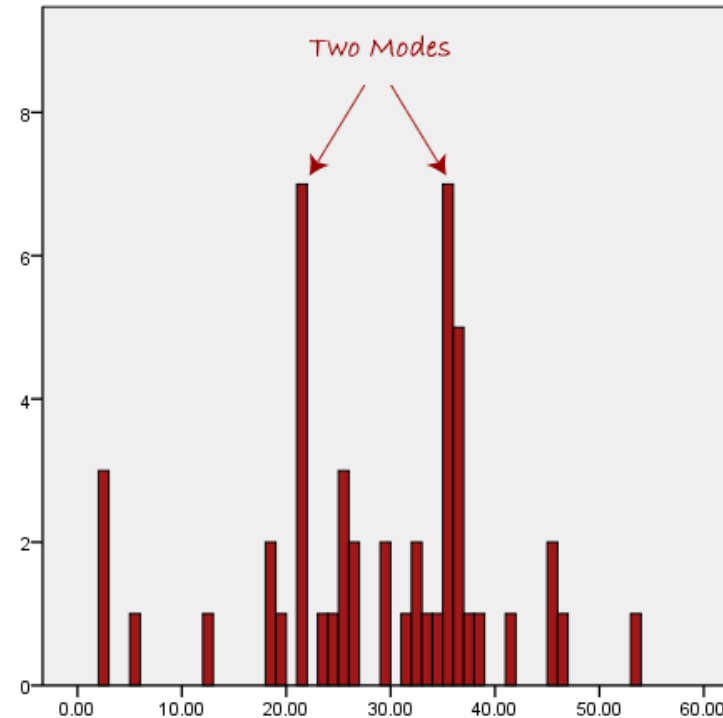
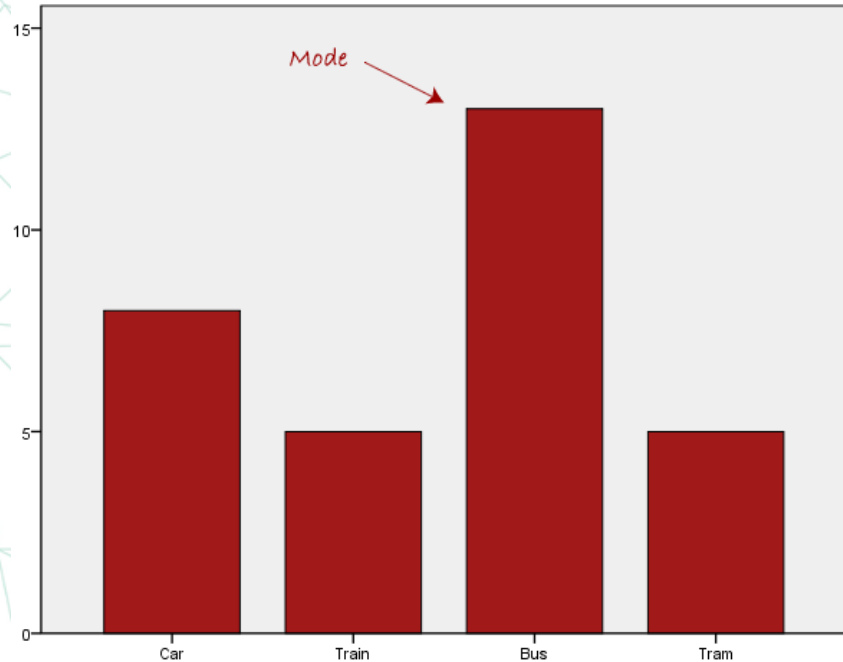
Median = $(4 + 5) \div 2$

= **4.5**

Mode

- **Mode** is the most frequent value or item in the dataset.
- A dataset can generally have **one** or **more than one** mode value.

Mode = Highest Frequency Term



Mode

The mode is the value that occurs most often. A dataset can have 0 modes, 1 mode or multiple modes.

The mode is calculated simply by finding the value with the highest frequency.

- **Importance:** The mode indicates the most frequently occurring value in a dataset, useful for understanding the most common occurrences.
- **Application:** Common in market research, quality control, and inventory management.

Example Data 1 [10, 13, 15, 16, 13, 15, 11, 13]

Example Data 2 [8, 9, 13, 11, 12, 16, 8, 10]

For data 1: 13 is the most frequent

For data 2: 8 is the most frequent

Applicant	Loan Amount
Jitesh	\$ 24,000
John	\$ 18,000
Frans	\$ 34,000
Danny	\$ 40,000
Cecile	\$ 24,000
Scott	\$ 16,000
Alex	\$ 29,000

Mode = \$24,000

Differences between Mean, Median and Mode

Feature	Mean	Median	Mode
Definition	Mean is the average of all values.	Median is the middle value when data is sorted.	Mode is the most frequently occurring value in the dataset.
Sensitivity	Mean is sensitive to outliers.	Median is not sensitive to outliers .	Mode is not sensitive to outliers.
Calculation	Calculated by adding up all values of a dataset and dividing them by the total number of values in dataset.	Calculated by finding the middle value in a list of data.	Calculated by finding which value occurs more number of times in a dataset.
Representation	Value of mean may or may not be in dataset.	Value of median is always a value from the dataset.	Value of mode is also always a value from the dataset.

Effect of Outliers in Descriptive Statistics

- An outlier is any unusually **large** or **small** observation.
- Outliers can have a disproportionate effect on statistical results, such as the mean but doesn't affect median or mode which can result in **misleading impressions**

Experience	Salary
1	\$ 3,725
2	\$ 4,155
3	\$ 4,627
4	\$ 5,147
5	\$ 5,718
6	\$ 6,347
7	\$ 7,039
8	\$ 7,210
9	\$ 7,423
10	\$ 7,556
11	\$ 8,369
12	\$ 8,810
13	\$ 8,940
14	\$ 9,200
15	\$ 9,458

6,915 \$ ← Mean → 7,676 \$

7,200 \$ ← Median → 7,200 \$

Experience	Salary
1	\$ 3,725
2	\$ 4,155
3	\$ 4,627
4	\$ 5,147
5	\$ 5,718
6	\$ 6,347
7	\$ 7,039
8	\$ 7,210
9	\$ 7,423
10	\$ 19,000
11	\$ 8,369
12	\$ 8,810
13	\$ 8,940
14	\$ 9,200
15	\$ 9,458

Effect of Outliers in Descriptive Statistics



Experience	Salary
1	\$ 3,725
2	\$ 4,155
3	\$ 4,627
4	\$ 5,147
5	\$ 5,718
6	\$ 6,347
7	\$ 7,039
8	\$ 7,210
9	\$ 7,423
10	\$ 19,000
11	\$ 8,369
12	\$ 8,810
13	\$ 8,940
14	\$ 9,200
15	\$ 9,458

Descriptive Statistics

Measure of Dispersion

Difference between Central Tendency and Dispersion

- **Central tendency** tells you where most of your data points lie, while **dispersion** summarizes how far apart your points are from each other.
- Datasets can have the same **central tendency** but different levels of **dispersion**, or vice versa. Together, they give you a complete picture of your data.



Central Tendency



Spread in data

- This example show how central tendency alone is not enough to describe data here we have same mean and median but each time with different data, so we need to measure dispersion

Day	Temperature
1	20
2	21
3	19
4	20
5	21
6	19
7	20
Total	140

Mean : 20

Median : 20

Day	Temperature
1	22
2	23
3	21
4	18
5	19
6	17
7	20
Total	140

Mean : 20

Median : 20

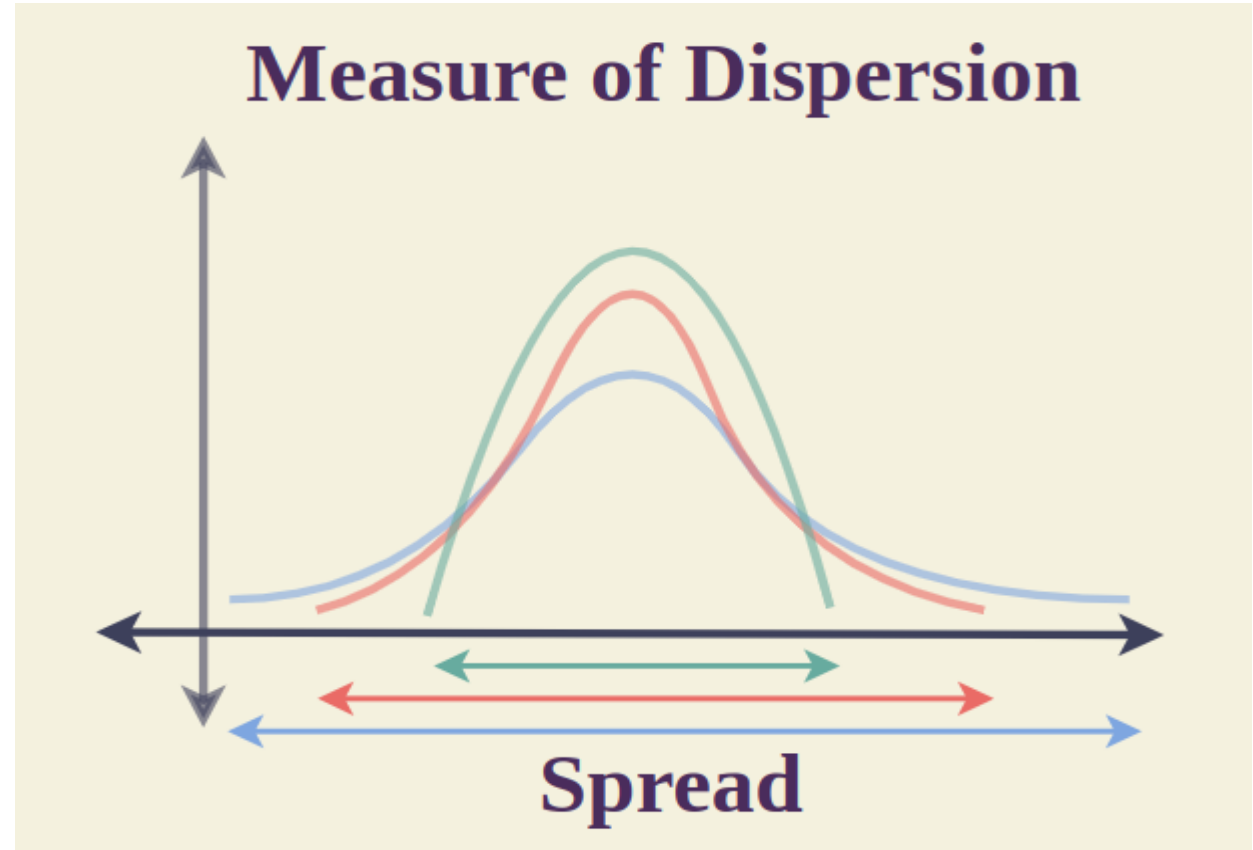
Day	Temperature
1	12
2	11
3	13
4	20
5	24
6	29
7	31
Total	140

Mean : 20

Median : 20

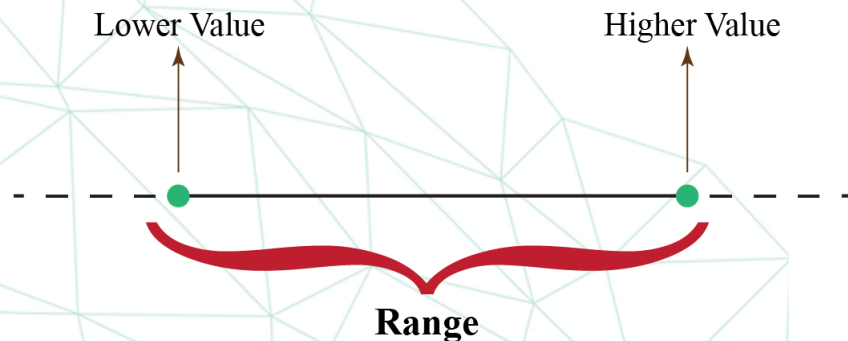
Measure of Dispersion

- Range
- Interquartile range
- Percentile
- Variance
- Standard deviation



Range

- **The range** is the difference between the highest value and the lowest value of the **data**. It helps in knowing the **spread** of the data.
- **Application:** Widely used in everyday scenarios such as evaluating temperature variations over a week, comparing prices of products, and understanding the performance spread of students in a test.



16, 24, 22, 25, 26, 27, 28, 23

$$\text{Range} = \text{max} - \text{min}$$

$$\text{Range} = 28 - 16 = 12$$

Question

Calculate the range of the given set of data:
7, 47, 8, 42, 47, 95, 42, 96, 2.

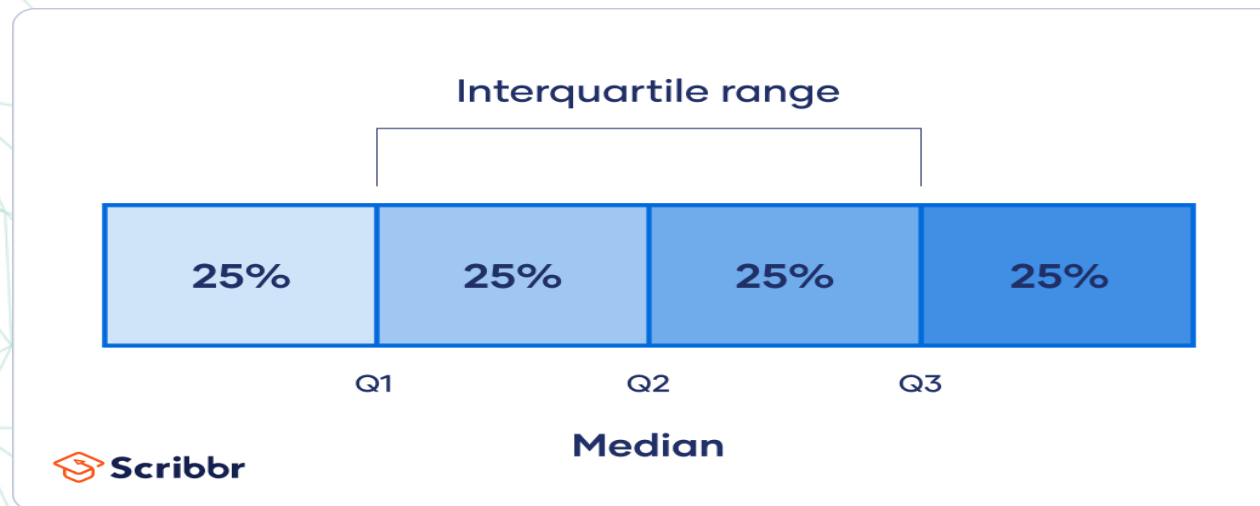
- A) 90
- B) 94
- C) 96
- D) 100

Range and Interquartile Range

- **It** is a better measure of dispersion than **range** because it leaves out the extreme values.
- It equally divides the distribution into four equal parts called **quartiles**:
 - The first 25% is the **1st quartile (Q1)**.
 - The middle one is the **2nd quartile (Q2)**.
 - The last one is the **3rd quartile (Q3)**.
- The **2nd quartile (Q2)** divides the distribution into two equal parts of 50%, so it is the same as the **Median**.
- The **interquartile range** is the distance between the third and the first quartile, or, in other words, **$IQR = Q3 - Q1$** .

Range and Interquartile Range

- The **interquartile range** is a measure of where the “middle fifty” is in a data set.
- Where a **range** is a measure of where the beginning and end are in a set, the **interquartile range** is a measure of where the bulk of the values lie.
- That’s why it’s preferred over many other measures of spread when reporting things like school performance or SAT scores.
- The **interquartile range formula** is the first quartile subtracted from the third quartile:
 - $IQR = Q3 - Q1$.



Quartile Calculations

Interquartile range (IQR) method

Calculate IQR:

To calculate the IQR, you can follow these steps:

- **Sort** the data in **ascending** order.
- Find the **median** of the dataset.
- Divide the dataset into **two halves**, the lower half, and the upper half. If the median is included in either half, include it in **both halves**.
- Find the **median** of the **lower** half (**Q1**) and the **median** of the **upper** half (**Q3**).
- Calculate the **IQR** by subtracting **Q1** from **Q3**: $IQR = Q3 - Q1$.

Example:

9, 3, 2, 5, 7, 8, 6, 1, 4, 10

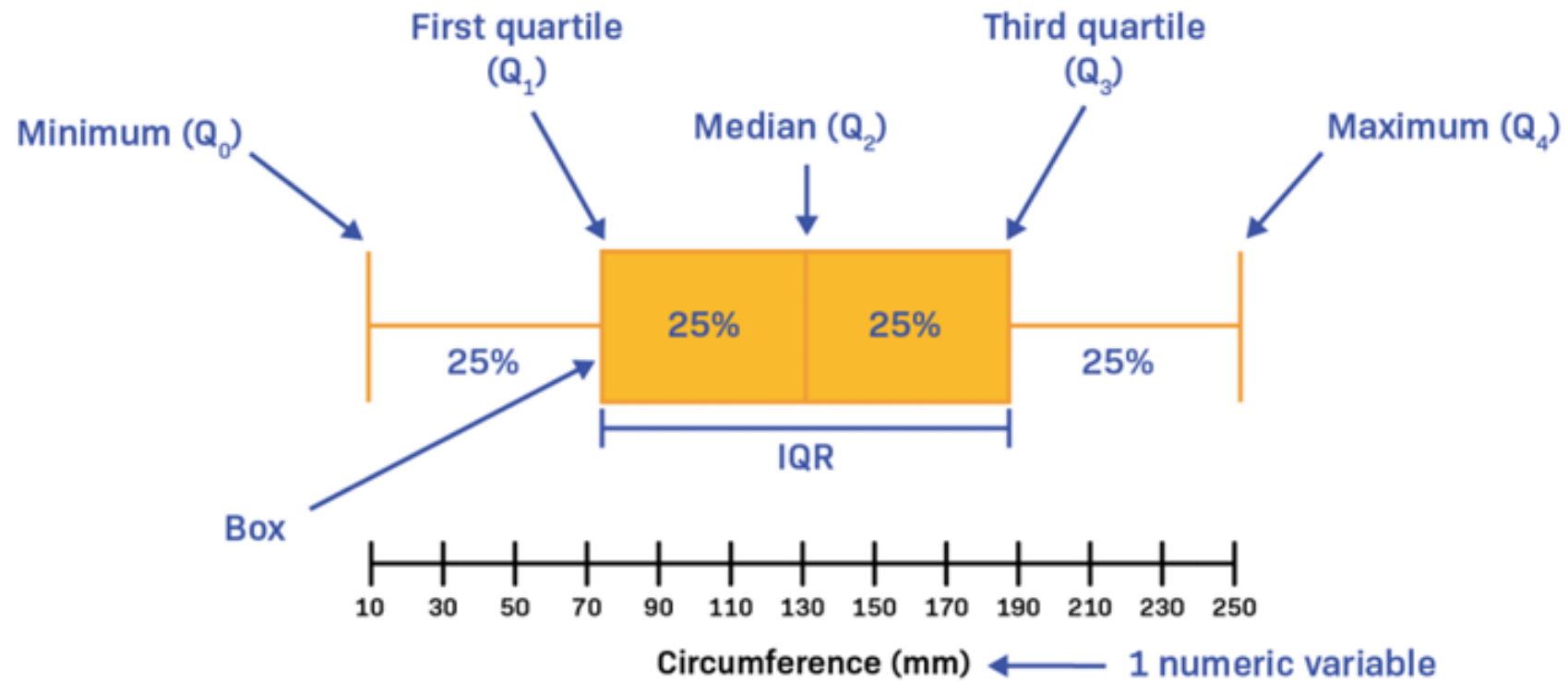
1, 2, 3, 4, **5, 6**, 7, 8, 9, 10 $\rightarrow$ **Q2** or **median** = $(5+6)/2 = 5.5$

Lower half: 1, 2, **3**, 4, 5 $\rightarrow$ **Q1** = **3**

Upper half: 6, 7, **8**, 9, 10 $\rightarrow$ **Q3** = **8**

IQR = **Q3** - **Q1** = $8 - 3 = 5$

Box Plot



Question

Calculate the interquartile range (IQR) of the following data:
17, 18, 18, 19, 20, 21, 21, 23, 25.

A) 4

B) 5

C) 6

D) 7

Question

Find the interquartile range (IQR) using the following values:

- Minimum: 1
- Q1: 3
- Median: 5
- Q3: 7
- Maximum: 9

A) 2

B) 3

C) 4

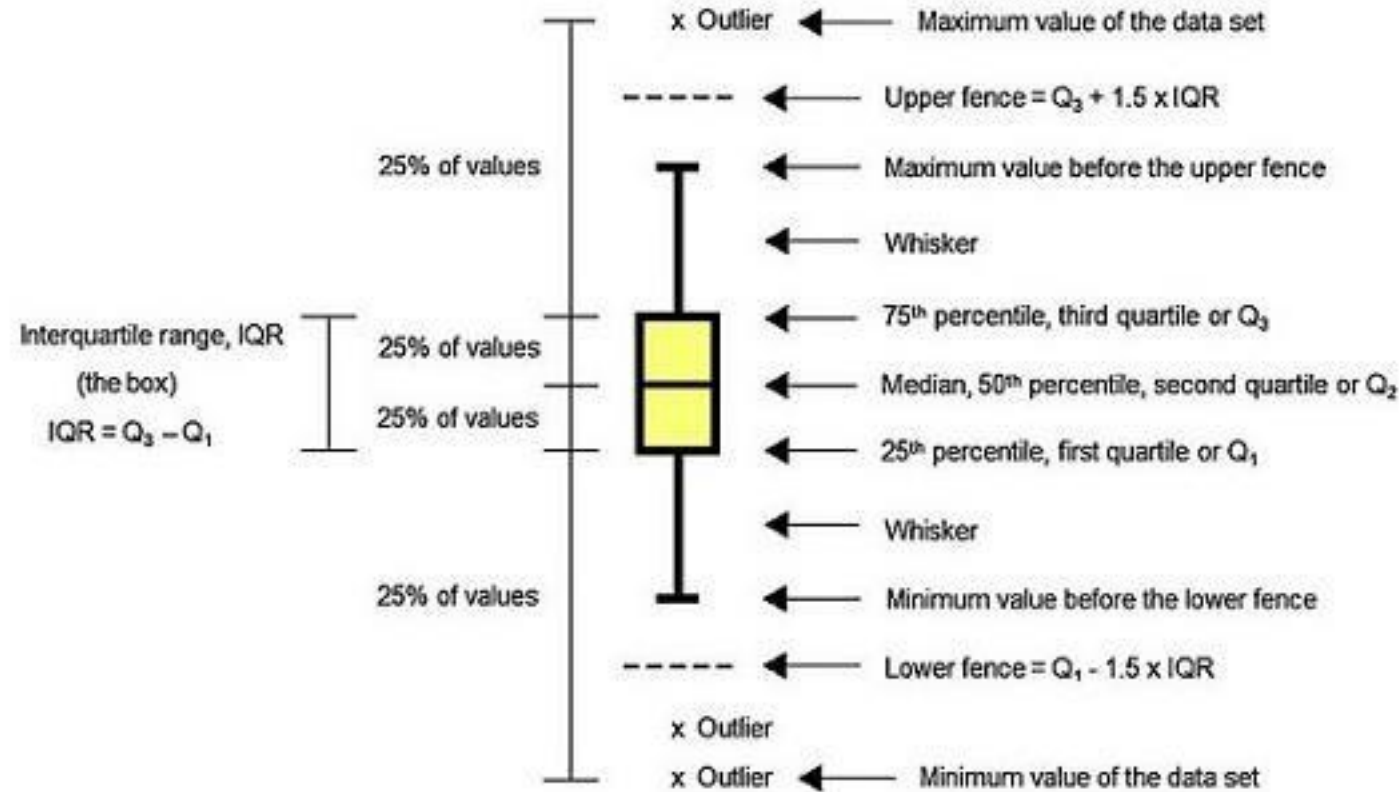
D) 5

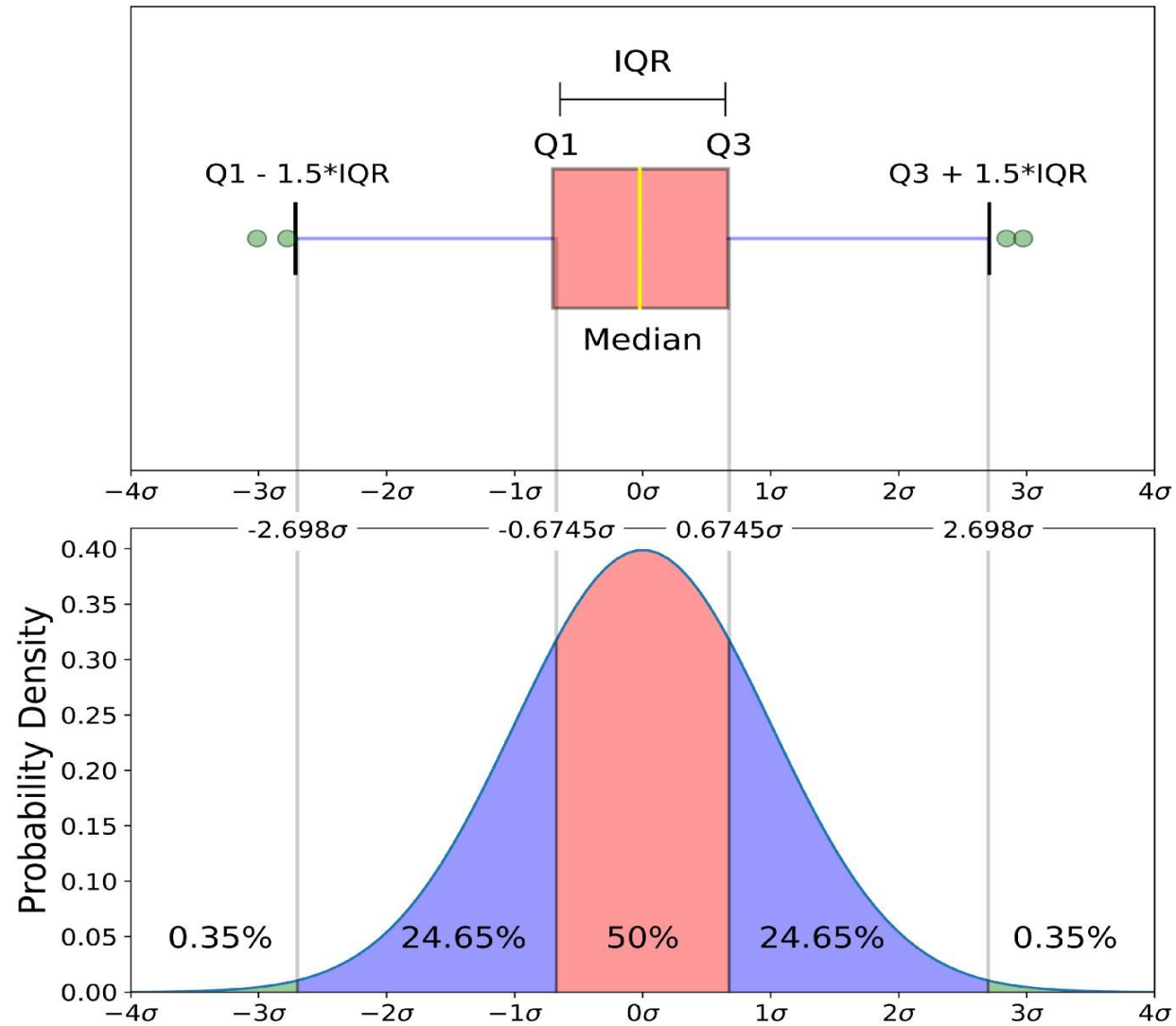
Advantage of IQR

- The main advantage of the **IQR** is that it is not affected by **outliers** because it doesn't consider observations below **Q1** or above **Q3**.
- Observations can be considered outliers when they lie more than **1.5 IQR** below the first quartile or **1.5 IQR** above the third quartile.
- **Outliers = Q1 – 1.5 × IQR (or) Q3 + 1.5 × IQR**

Box Plot

- mainly used when you are describing center and variability of your data.
- It is also useful for detecting outliers in the data.
- Used to visualize IQR





IQR is plotted in a boxplot and probability density

Percentile

- A percentile is a measure used in statistics to indicate the value below which a given percentage of observations in a group of observations fall.

$$\text{Percentile}(x) = (\text{Number of values fall under 'x'}/\text{total number of values}) \times 100$$

$$P = (n/N) \times 100$$

Where ,

- P is percentile
- n – Number of values below 'x'
- N – Total count of population

Percentile calculation

- The number that expresses the value that a given percent of the values are lower than
- **Example: (For Calculating a percentile for a value)**
- We have an array of the ages of all the people that working in same office ages = [25,31,43,48,50,41,39,60,52,32,27,46,47,55]
- Calculate the percentile for the value 48
 - Arrange in ascending order [25,27,31,32,39,41,43,46,47,48,50,52,55,60]
 - Find the number of values less than 48 , $n = 9$, $N = 14$
 - Percentile = $n/N \times 100 = (9/14) \times 100 = 64.28\%$
 - Then 64% of the ages are less than 48

Percentile calculation

- The number that expresses the value that a given percent of the values are lower than
- **Example: (For Calculating a percentile for a value)**
- We have an array of the ages of all the people that working in same office ages = [25,31,43,48,50,41,39,60,52,32,27,46,47,55]
- Calculate the 64th percentile for the value 48
 - Arrange in ascending order [25,27,31,32,39,41,43,46,47,48,50,52,55,60]
 - $L_p = (N + 1) \frac{P}{100}$
 - $L_{64} = (14 + 1) * \frac{64}{100} = 9.6$
 - Round to 10 the and count the tenth value it will be 48

Percentile

- **Arrange** the data in an order
- **Calculate** the **percentage** of observations or data points below a particular value.

What is the **80th** Percentile observation?

Total Observations * 0.8

$$15 * 0.8 = \underline{12}$$

Row Number	Salary
1	\$ 3,725
2	\$ 4,155
3	\$ 4,627
4	\$ 5,147
5	\$ 5,718
6	\$ 6,347
7	\$ 7,039
8	\$ 7,210
9	\$ 7,423
10	\$ 7,556
11	\$ 8,369
12	\$ 8,810
13	\$ 8,940
14	\$ 9,200
15	\$ 9,458

Variance

- **Variance** is a measure of how far a set of data are dispersed from their **mean** or **average** value. It is denoted as ' σ^2 '.

Properties of Variance:

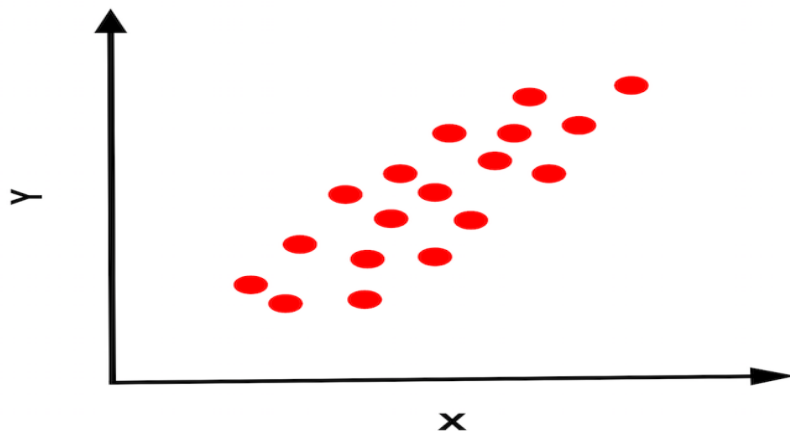
- It is always non-negative since each term in the variance sum is squared, and therefore, the result is either positive or zero.
- Variance always has **squared units**. For example, the variance of a set of weights estimated in kilograms will be given in **kg²**.
- Since the population variance is squared, we cannot compare it directly with the **mean** or the data themselves.

Variance (σ^2)

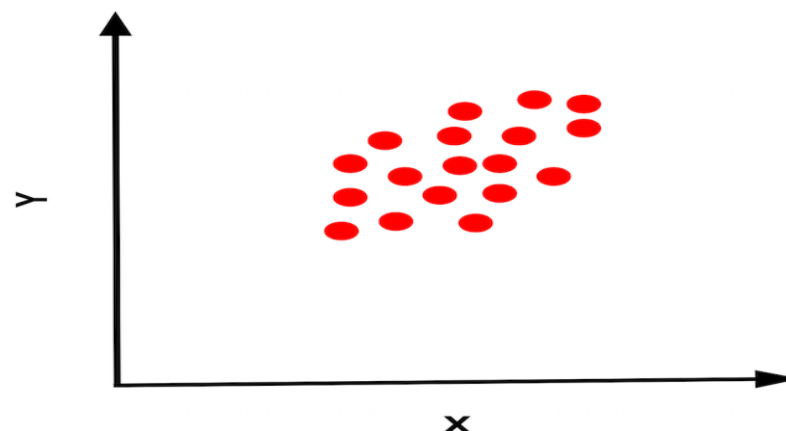
Variance measures the dispersion of a set of data points around its mean value.

- **Importance:** Variance measures the spread of data points around the mean, useful for understanding data variability.
- **Application:** Used in risk management, investment analysis, and quality control.

In Figure 1, the points have a high variance because they are spread out.



In Figure 2, the points have a low variance because they are close together.



How to Calculate Variance Step by Step

- **Calculate the Mean ($\bar{x}$):** Find the average of all data points.
- **Subtract the Mean from Each Observation ($X - \bar{x}$):** For each data point, subtract the mean from the data point.
- **Square Each of the Resulting Observations ($(X - \bar{x})^2$):** Square the result of each subtraction.
- **Add These Squared Results Together:** Sum all the squared values.
- **Divide This Total by the Number of Observations (n)** (in the case of a population) to Get **Variance (σ^2)**: Divide the sum of squared values by the number of observations to obtain the variance.

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

Variance Example

Day	X	$X - \bar{X}$	$(X - \bar{X})^2$
1	20	0	0
2	21	1	1
3	19	-1	1
4	20	0	0
5	21	1	1
6	19	-1	1
7	20	0	0

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

- Average ($\bar{x}$) = $140/7 = 20$
- σ^2 = Variance = $4/7 = 0.57$

Standard Deviation

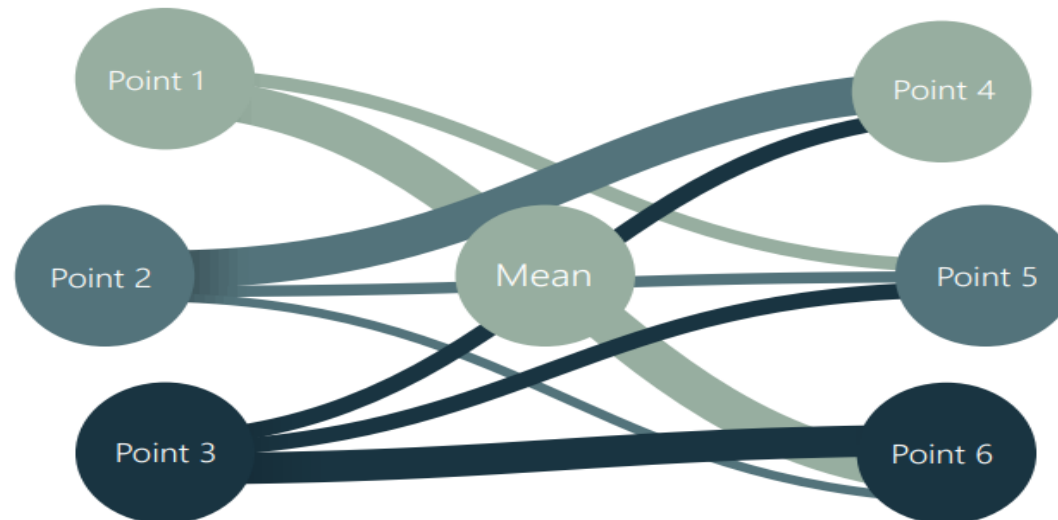
- **Standard deviation** measures the deviation of data from its mean or average position. The degree of dispersion is computed by estimating the deviation of data points. It is denoted by the symbol ' σ '.
- **Properties of Standard Deviation:**
 - It describes the square root of the mean of the squares of all values in a data set and is also called the **root-mean-square deviation**.
 - The smallest value of the standard deviation is **0** since it cannot be negative.
 - When the data values of a group are similar, the standard deviation will be very low or close to zero. However, when the data values vary significantly from each other, the standard deviation will be high or farther from zero.

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

Standard deviation std (σ)

The std is the most common way to measure the spread of the data and how close data points are to each other.

- **Importance:** Standard deviation is the square root of variance, providing a measure of data dispersion that is in the same unit as the data. $\sigma = \sqrt{\sigma^2}$
- **Application:** Commonly used in finance for assessing investment risk, and in process control for monitoring production quality.



Variance and standard deviation

Day	Temperature
1	20
2	21
3	19
4	20
5	21
6	19
7	20

$$\sigma = 0.7559$$

$$\text{Mean} = \bar{X} = 20$$

Day	Temperature
1	12
2	11
3	13
4	20
5	24
6	29
7	31

$$\sigma = 7.67$$

$$\text{Mean} = \bar{X} = 20$$

Task

Calculate Mean ,Median ,Mode ,Variance ,Std for these students in Math and Science Subjects Grades

Student	Math	Science
A	85	78
B	90	88
C	78	84
D	92	91
E	88	76

Task Solution

Calculate Mean ,Median ,Mode ,Variance ,Std for these students in Math and Science Subjects Grades

Math

$$\text{Mean} = (85+90+78+92+88)/5 = 86.6$$

$$\text{Median} = [78, 85, 88, 90, 92]$$

Mode = No mode (all grades are unique)

$$\text{Variance} = 23.84$$

$$\text{Std} = 4.88$$

Because

$$85 - 86.6 = -1.6 \quad (-1.6)^2 = 2.56$$

$$90 - 86.6 = 3.4 \quad (3.4)^2 = 11.56 \quad \text{variance} = (2.56 + 11.56 + 73.96 + 29.16 + 1.96)/5 = 23.84$$

$$78 - 86.6 = -8.6 \quad (-8.6)^2 = 73.96 \quad \text{Std} = \sqrt{23.84} = 4.88$$

$$92 - 86.6 = 5.4 \quad (5.4)^2 = 29.16$$

$$88 - 86.6 = 1.96 \quad (1.96)^2 = 1.96$$

Task Solution

Calculate Mean ,Median ,Mode ,Variance ,Std for these students in Math and Science Subjects Grades

Science

$$\text{Mean} = (78+88+84+91+76)/5 = 83.4$$

$$\text{Median} = [76,78,84,88,91]$$

Mode = No mode (all grades are unique)

$$\text{Variance} = 32.64$$

$$\text{Std} = 5.71$$

Because

$$78 - 83.4 = -5.4 \quad (-5.4)^2 = 29.16$$

$$88 - 83.4 = 4.6 \quad (4.6)^2 = 21.16 \quad \text{variance} = (29.16 + 21.16 + 0.36 + 57.76 + 54.76)/5 = 32.64$$

$$84 - 83.4 = 0.6 \quad (0.6)^2 = 0.36 \quad \text{Std} = \sqrt{32.64} = 5.71$$

$$91 - 83.4 = 7.6 \quad (7.6)^2 = 57.76$$

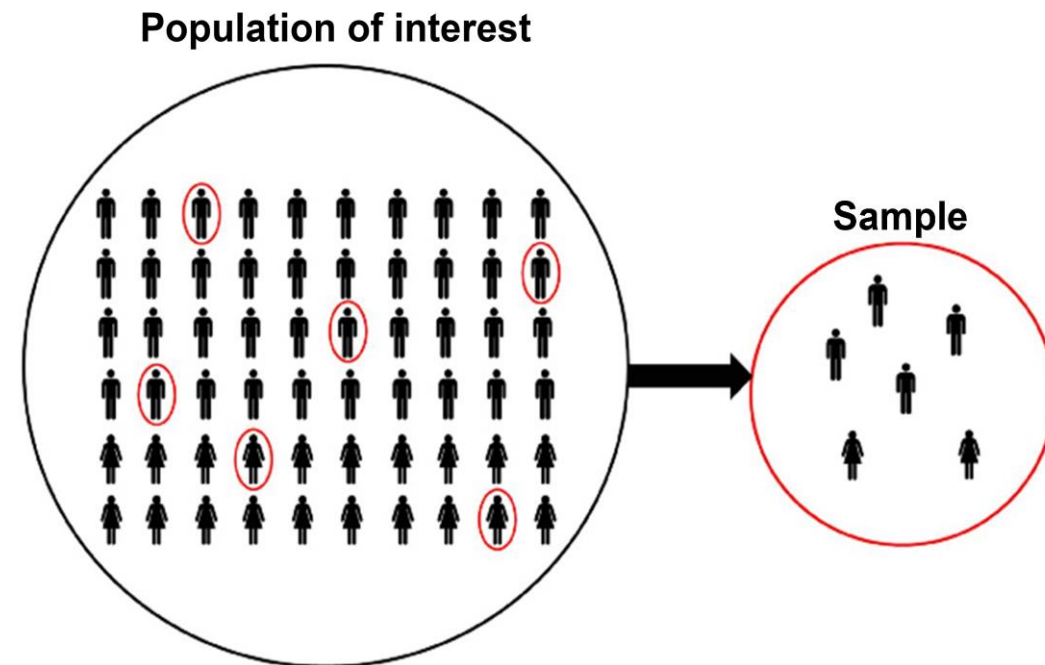
$$76 - 83.4 = -7.4 \quad (-7.4)^2 = 54.76$$

Inferential Statistics

Population VS. Sampling

Population Vs. Sampling

- **The primary task of inferential statistics** is making an inference about something by using only an incomplete sample of data.
- **Population:** is a collection of all possible individuals, objects, or measurements of interest (To understand the whole collection of data).
- **Sample:** is a portion or part of the population of interest (To make inferences about the whole based on the subset).



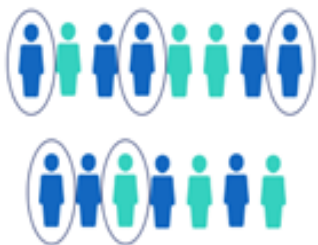
Sampling

Definition: The process of selecting a subset of individuals from a population to estimate characteristics of the whole population.

Types of Sampling:

- **Random Sampling:** Every member has an equal chance of being selected.
- **Stratified Sampling:** Population divided into subgroups, and samples are drawn from each.
- **Systematic Sampling:** Selecting every n th member from a list.
- **Convenience Sampling:** Selecting individuals who are easiest to reach, which may introduce bias.

Simple random sample



Systematic sample



Stratified sample



Cluster sample



When Should Samples be used?

- **When studying a large population** where it is impractical or impossible to collect data from every individual.
- **When resources** such as time, cost, and manpower are limited, making it more feasible to collect data from a subset of the population.
- **When conducting research or experiments** where it is important to minimize potential biases in data collection.

For samples:

$$\text{variance} = s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$\text{standard deviation} = s = \sqrt{s^2}$$

Calculating Formula

$$s^2 = \frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n - 1}$$

$$S^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

For populations:

$$\text{variance} = \sigma^2 = \frac{\sum (x - \bar{x})^2}{n}$$

$$\text{standard deviation} = \sigma = \sqrt{\sigma^2}$$

Calculating Formula

$$\sigma^2 = \frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n}$$

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

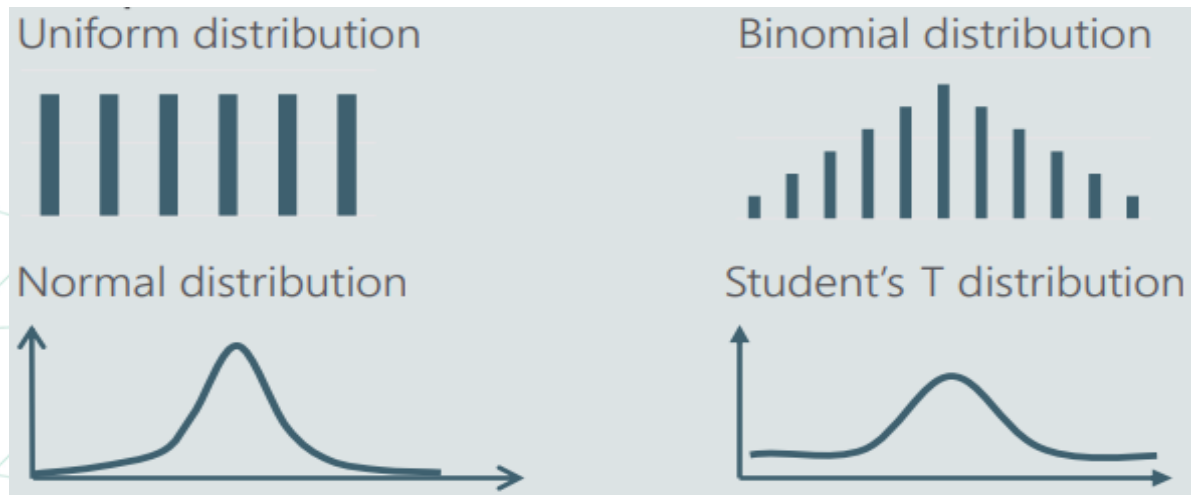
Data Distribution

- A distribution is a function that shows the possible values for a variable and how often they occur
- In probability theory and statistics, a probability distribution is a mathematical function that provides the probabilities of the occurrence of different possible outcomes in an experiment
- It is a common mistake to believe that the distribution is the graph. In fact, the distribution is the 'rule' that determines how values are positioned in relation to each other. Very often, we use a graph to visualize the data. Since different distributions have a particular graphical representation, statisticians like to plot them.

Probability Distribution

Common distributions include:

- **Normal Distribution:** Symmetrical, bell-shaped distribution characterized by its mean and standard deviation.
- **Binomial Distribution:** Represents the number of successes in a fixed number of trials with a constant probability of success.
- **Poisson Distribution:** Models the number of events occurring in a fixed interval of time or space.



Normal Distribution

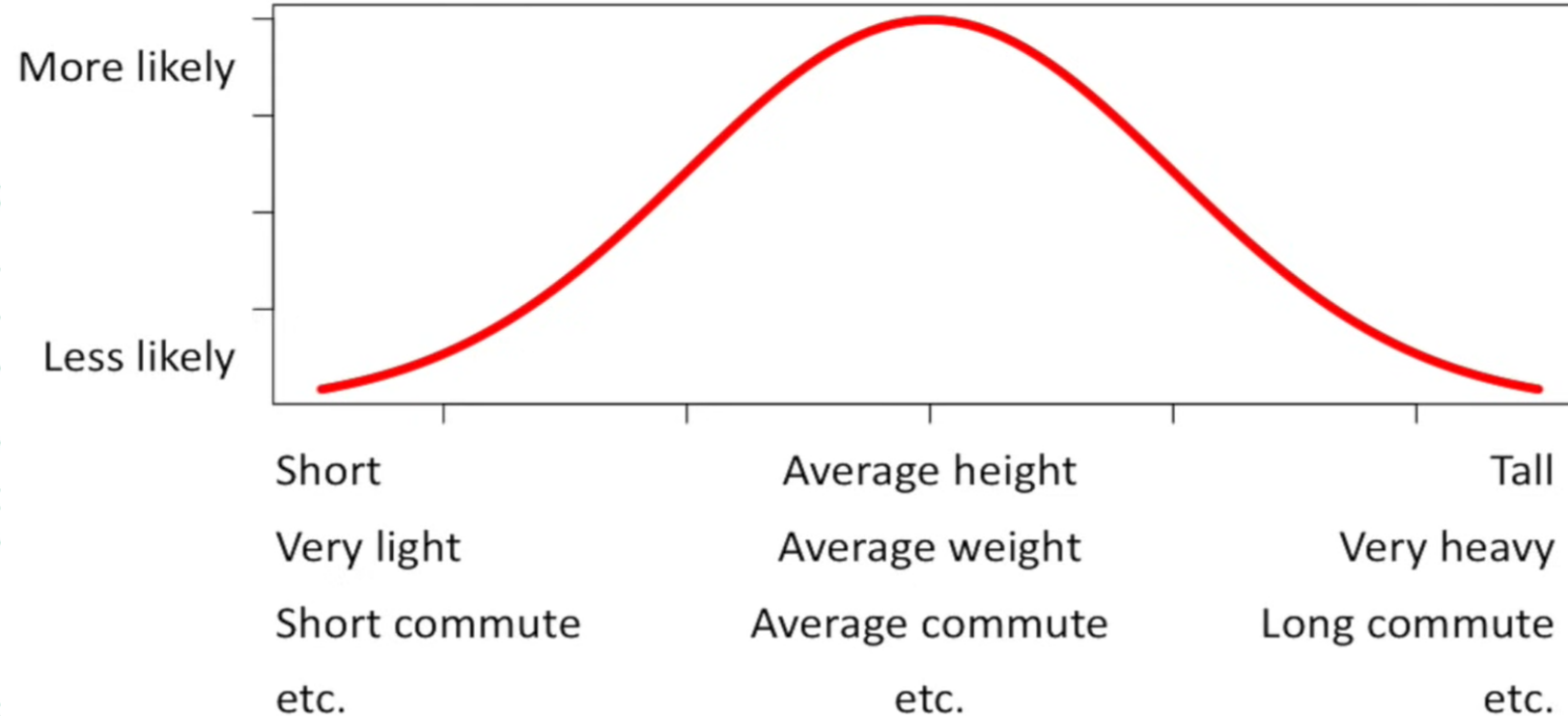
- ❑ **Normal distribution**, also known as the Gaussian distribution, is a probability distribution that is symmetric about the mean, showing that data near the mean are more frequent in occurrence than data far from the mean (Bell shape).

- ❑ **The Importance of Normal Distribution**
 1. It approximates a wide variety of random variables
 2. Distributions of sample means with large enough sample sizes could be approximated as normal
 3. All computable statistics are well-structured
 4. It is widely used in regression analysis
 5. Good track record

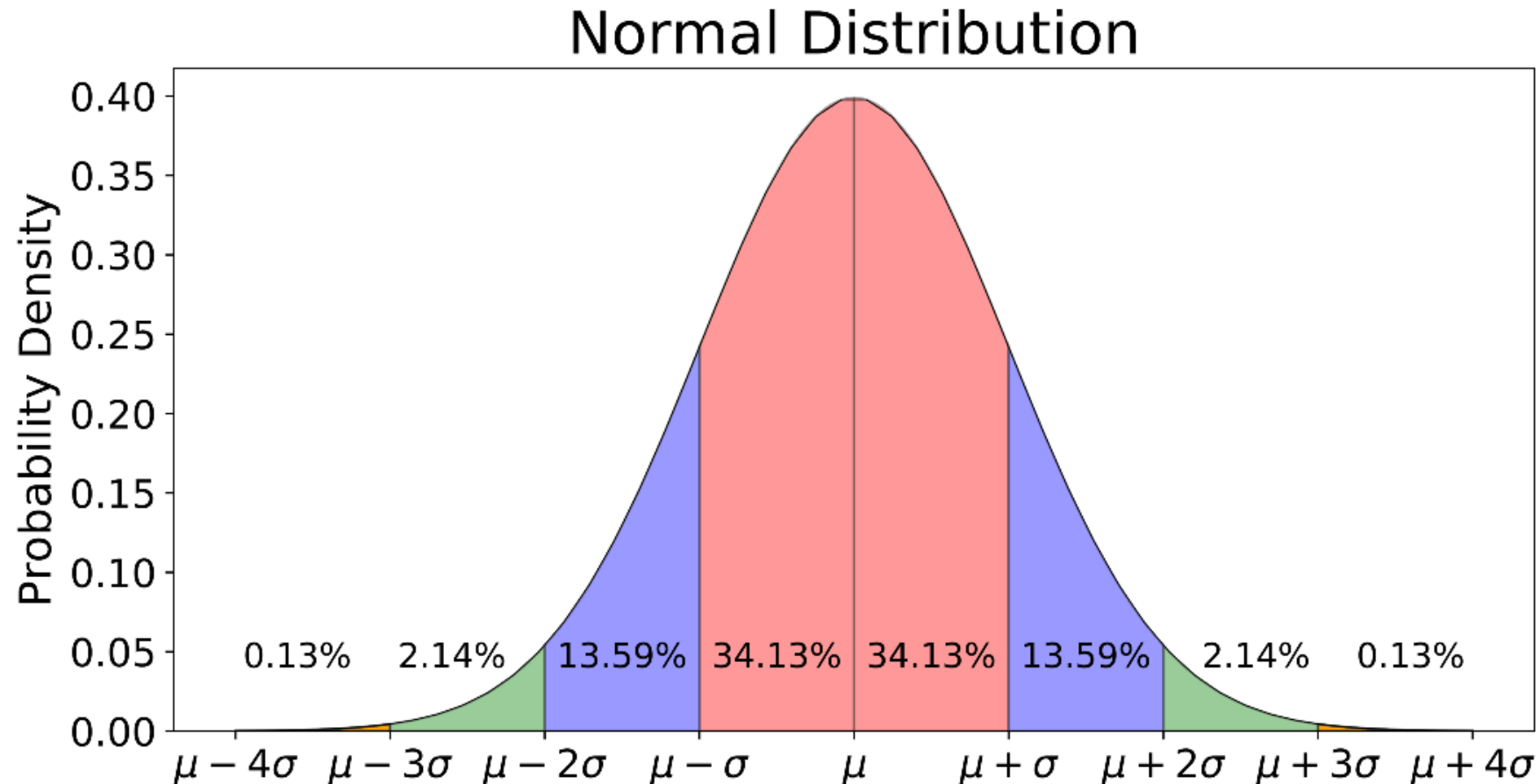
- ❑ **Applications:**
 - **Biology:** Most biological measures are normally distributed, such as: height, arm and leg length, nails, blood pressure, thickness of tree barks, etc.
 - **IQ tests, Stock Market Information**

Normal Distribution

A lot of things are “normally distributed”.



How is Data Distributed in Normal Distribution?



What is Skewness of Data?

- Skewness is the measure of how much the probability distribution of a random variable deviates from the normal distribution.
- Skewness is positive if the tail of the distribution extends more to the right, and negative if the tail extends more to the left.

Mean	Median	Mode
2.79	2.00	2.00

Positive skew



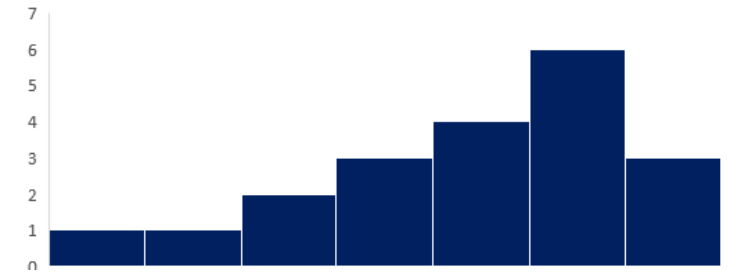
Mean	Median	Mode
4.00	4.00	4.00

Zero skew

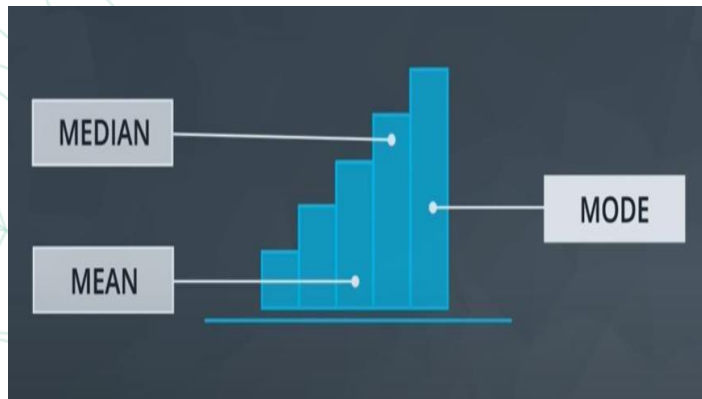
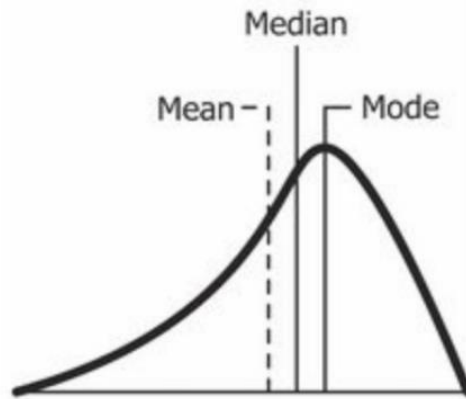


Mean	Median	Mode
4.90	5.00	6.00

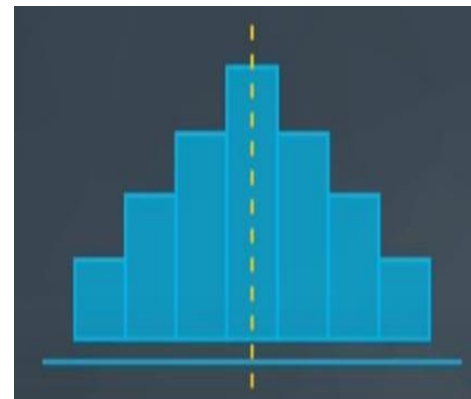
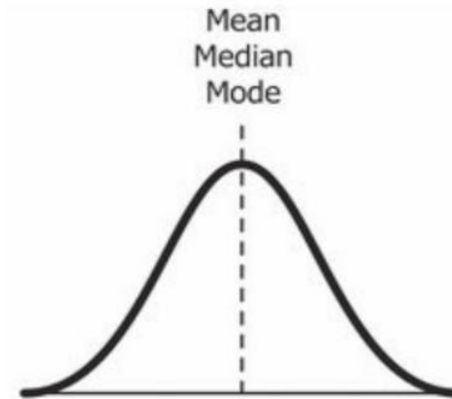
Negative skew



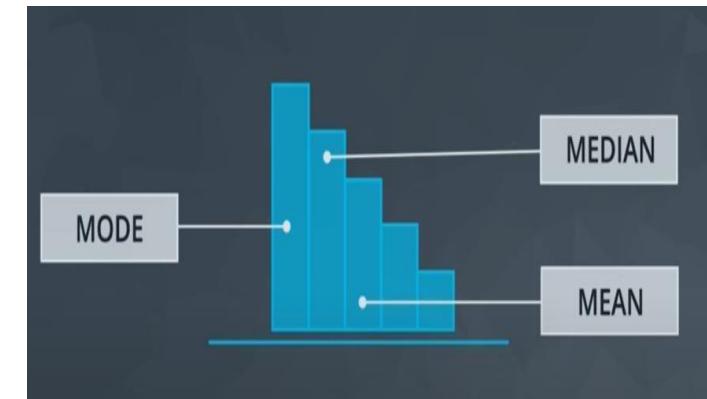
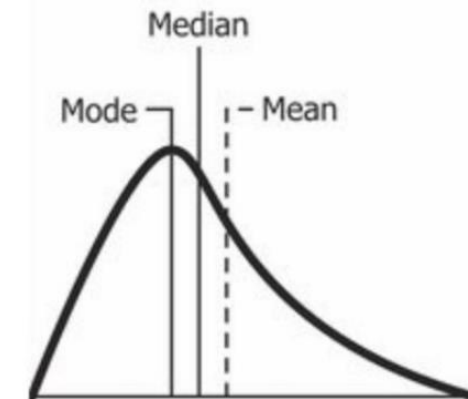
Mean, Median and Mode



The value of skewness for a **left skewed distribution** is less than zero.
 $\text{mean} < \text{median} < \text{mode}$



The value of skewness for a **normal distribution** is zero.
 $\text{mode} = \text{median} = \text{mean}$



The value of skewness for a **right skewed distribution** is less than zero
 $\text{mode} < \text{median} < \text{mean}$

Correlation

- ❑ Used to find the relationship between two variables which is important in real because we can predict value of one variable.

Example from our daily life:

- ❑ The more time you spend running on a treadmill, the more calories you will burn.
- ❑ The more money you save, the more financially secure you feel.
- ❑ The more cigarettes you smoked, the higher stress level you have

Use Scatter plot in visualization of correlation

Correlation Coefficient

- ❑ The correlation coefficient (r)/pearson r indicates a measure of the strength of a relationship between two variables.
Possible correlations range from $+1$ to -1 .

Direction of correlation :

- A correlation of -1 indicates a perfect negative correlation, meaning that as one variable goes up, the other goes down.
- A correlation of $+1$ indicates a perfect positive correlation , meaning that as one variable goes up, the other goes up together.
- A correlation of zero indicates that there is no relationship between the variables.

- ❑ Describe linear relations only

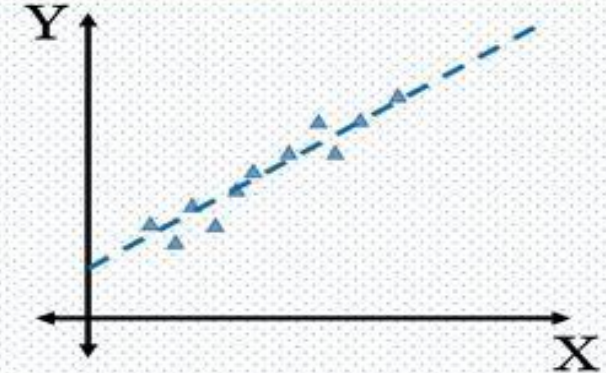
Correlation Coefficient

Statistically Correlated

- Strength of the correlation – Coefficient of Correlation
- Direction of correlation – Sign of the Coefficient

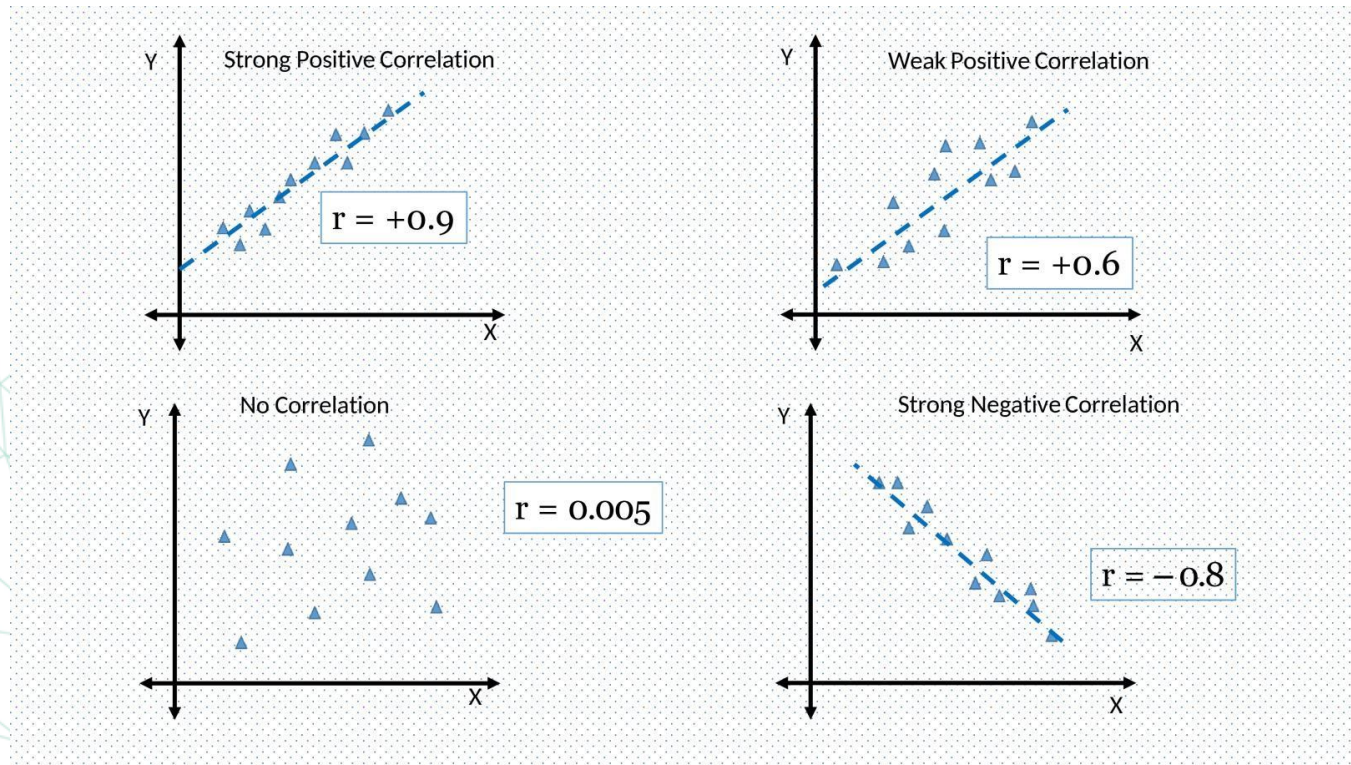
Pearson Correlation
Coefficient

$$r = \frac{\sum (x - \bar{x}) * (y - \bar{y})}{(N - 1) * \sigma_x * \sigma_y}$$

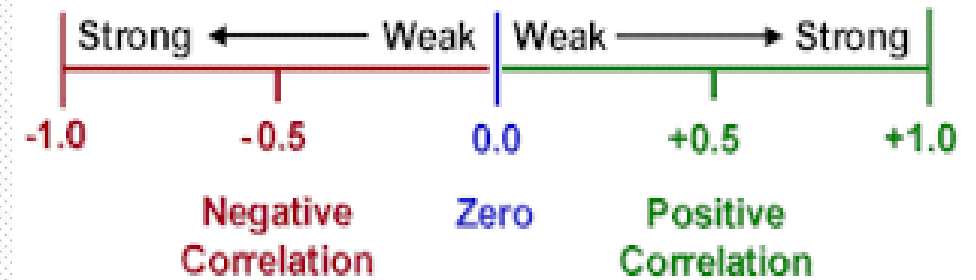


- **X**: Values of the **x-variable** in a sample
- **$\bar{x}$** : Mean of the values of the **x-variable**
- **y**: Values of the **y-variable** in a sample
- **$\bar{y}$** : Mean of the values of the **y-variable**
- **N**: Number of records
- **σ** : Standard deviation

Correlation Coefficient



Correlation Coefficient
Shows Strength & Direction of Correlation



Scatter plot in visualization of correlation , The horizontal axis represents one variable, and the vertical axis represents the other.

Regression line : is the line separate the 2 classes As points surrounded regression line the strength of correlation increase

The closer the correlation is to 0, the weaker it is, while the closer it is to +/-1, the stronger it is.

Correlation Coefficient

	Height X	Weight Y	$X - \bar{X}$	$Y - \bar{Y}$	$(X - \bar{X}) * (Y - \bar{Y})$
	160	130	-15.625	-40.625	634.7656
	170	150	-5.625	-20.625	116.0156
	165	145	-10.625	-25.625	272.2656
	180	190	4.375	19.375	84.76563
	175	175	-0.625	4.375	-2.73438
	190	210	14.375	39.375	566.0156
	185	180	9.375	9.375	87.89063
	180	185	4.375	14.375	62.89063
Mean	175.625	170.625			1821.875
Std Dev	10.155	25.651			

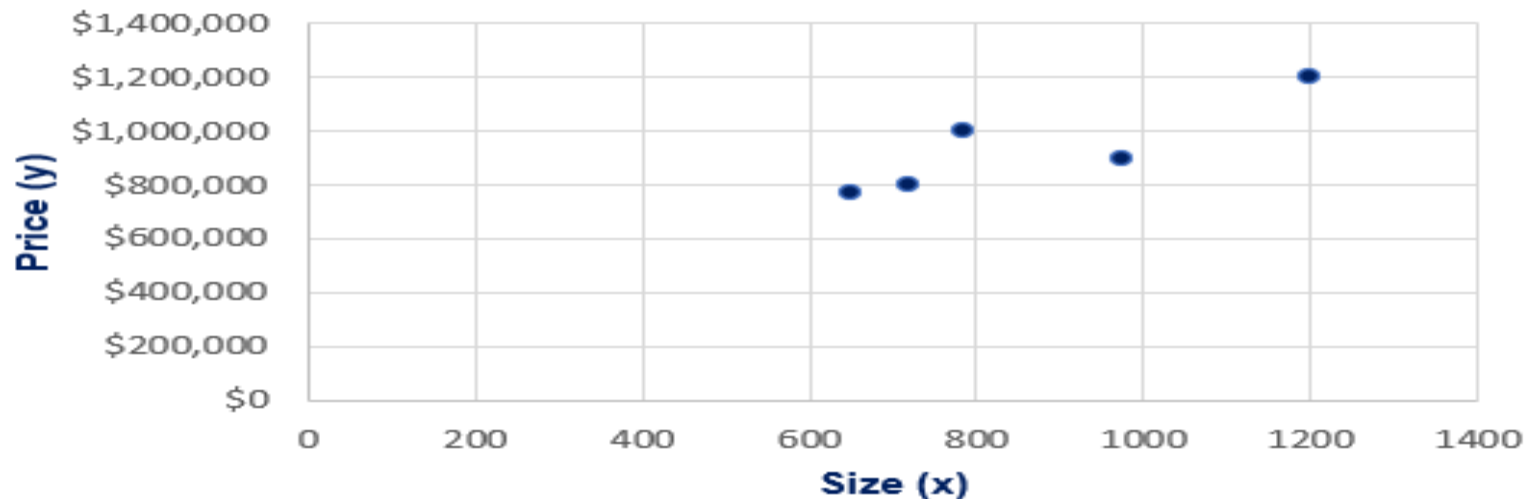
$$r = \frac{\sum (x - \bar{x}) * (y - \bar{y})}{(N - 1) * \sigma_x * \sigma_y}$$

$$r = \frac{1821.875}{(8-1) * 10.155 * 25.651}$$

$$r = 0.96$$

Example: Housing Data

	Size (ft.)	Price (\$)	$(x-\bar{x})*(y-\bar{y})$
	650	772,000	34,776,000
	785	998,000	- 5,265,000
	1200	1,200,000	89,178,000
	720	800,000	19,418,000
	975	895,000	- 4,142,000
Mean	866	933,000	
Sum			133,965,000
Sample size			5
Cov. Sample			33,491,250
Corr Coefficient			0.87



Sampling “Error

SAMPLING ERROR The difference between a sample statistic and its corresponding population parameter.

Example

Refer to the previous example on page 268, where we studied the number of rooms rented at the Foxtrot Inn bed and breakfast in Tryon, North Carolina. The population is the number of rooms rented each of the 30 days in June 2011. Find the mean of the population. Use Excel or other statistical software to select three random samples of five days. Calculate the mean of each sample and compare it to the population mean. What is the sampling error in each case?

Solution

During the month, there were a total of 94 rentals. So the mean number of units rented per night is 3.13. This is the population mean. Hence we designate this value with the Greek letter μ .

$$\mu = \frac{\sum X}{N} = \frac{0 + 2 + 3 + \cdots + 3}{30} = \frac{94}{30} = 3.13$$

The first random sample of five nights resulted in the following number of rooms rented: 4, 7, 4, 3, and 1. The mean of this sample of five nights is 3.8 rooms, which we designate as $\bar{X}_1$. The bar over the X reminds us that it is a sample mean and the subscript 1 indicates it is the mean of the first sample.

$$\bar{X}_1 = \frac{\sum X}{n} = \frac{4 + 7 + 4 + 3 + 1}{5} = \frac{19}{5} = 3.80$$

Sampling “Error

In the third random sample, the mean was 1.8 and the sampling error was -1.33 .

Each of these differences, 0.67, 0.27, and -1.33 , is the sampling error made in estimating the population mean. Sometimes these errors are positive values, indicating that the sample mean overestimated the population mean; other times they are negative values, indicating the sample mean was less than the population mean.

Random Samples								
	A	B	C	D	E	F	G	H
1	June	Rentals			Sample 1	Sample 2	Sample 3	
2	1	0			4	3	0	
3	2	2			7	3	0	
4	3	3			4	2	3	
5	4	2			3	3	3	
6	5	3			1	6	3	
7	6	4		Totals	19	17	9	
8	7	2		Sample means	3.8	3.4	1.8	
9	8	3						
10	9	4						
11	10	7						
12	11	3						
13	12	4						
14	13	4						
15	14	4						
16	15	7						

In this case, where we have a population of 30 values and samples of 5 values, there is a very large number of possible samples—142,506 to be exact! To find this value, use the combination formula (5-10) in Section 5.8 on page 174. Each of the 142,506 different samples has the same chance of being selected. Each sample may have a different sample mean and therefore a different sampling error. The value of the sampling error is based on the particular one of the 142,506 different possible samples selected. Therefore, the sampling errors are random and occur by chance. If you were to determine the sum of these sampling errors over a large number of samples, the result would be very close to zero. This is true because the sample mean is an *unbiased estimator* of the population mean.

A sample mean is an unbiased estimate of the population mean.

Sampling Distribution of the Sample Mean

SAMPLING DISTRIBUTION OF THE SAMPLE MEAN A probability distribution of all possible sample means of a given sample size.

Example

Tartus Industries has seven production employees (considered the population). The hourly earnings of each employee are given in Table 8-2.

TABLE 8-2 Hourly Earnings of the Production Employees of Tartus Industries

Employee	Hourly Earnings	Employee	Hourly Earnings
Joe	\$7	Jan	\$7
Sam	7	Art	8
Sue	8	Ted	9
Bob	8		

1. What is the population mean?
2. What is the sampling distribution of the sample mean for samples of size 2?
3. What is the mean of the sampling distribution?
4. What observations can be made about the population and the sampling distribution?

Solution

Here are the solutions to the questions.

1. The population mean is \$7.71, found by:

$$\mu = \frac{\sum X}{N} = \frac{\$7 + \$7 + \$8 + \$8 + \$7 + \$8 + \$9}{7} = \$7.71$$

Sampling Distribution of the Sample Mean

We identify the population mean with the Greek letter μ . Our policy, stated in Chapters 1, 3, and 4, is to identify population parameters with Greek letters.

- To arrive at the sampling distribution of the sample mean, we need to select all possible samples of 2 without replacement from the population, then compute the mean of each sample. There are 21 possible samples, found by using formula (5-10) in Section 5.8 on page 174.

$${}_N C_n = \frac{N!}{n!(N-n)!} = \frac{7!}{2!(7-2)!} = 21$$

where $N = 7$ is the number of items in the population and $n = 2$ is the number of items in the sample.

The 21 sample means from all possible samples of 2 that can be drawn from the population are shown in Table 8-3. These 21 sample means are used to construct a probability distribution. This is the sampling distribution of the sample mean, and it is summarized in Table 8-4.

TABLE 8-3 Sample Means for All Possible Samples of 2 Employees

Sample	Employees	Hourly Earnings	Sum	Mean	Sample	Employees	Hourly Earnings	Sum	Mean
1	Joe, Sam	\$7, \$7	\$14	\$7.00	12	Sue, Bob	\$8, \$8	\$16	\$8.00
2	Joe, Sue	7, 8	15	7.50	13	Sue, Jan	8, 7	15	7.50
3	Joe, Bob	7, 8	15	7.50	14	Sue, Art	8, 8	16	8.00
4	Joe, Jan	7, 7	14	7.00	15	Sue, Ted	8, 9	17	8.50
5	Joe, Art	7, 8	15	7.50	16	Bob, Jan	8, 7	15	7.50
6	Joe, Ted	7, 9	16	8.00	17	Bob, Art	8, 8	16	8.00
7	Sam, Sue	7, 8	15	7.50	18	Bob, Ted	8, 9	17	8.50
8	Sam, Bob	7, 8	15	7.50	19	Jan, Art	7, 8	15	7.50
9	Sam, Jan	7, 7	14	7.00	20	Jan, Ted	7, 9	16	8.00
10	Sam, Art	7, 8	15	7.50	21	Art, Ted	8, 9	17	8.50
11	Sam, Ted	7, 9	16	8.00					

Sampling Distribution of the Sample Mean

TABLE 8-4 Sampling Distribution of the Sample Mean for $n = 2$

Sample Mean	Number of Means	Probability
\$7.00	3	.1429
7.50	9	.4285
8.00	6	.2857
8.50	3	.1429
	21	1.0000

3. The mean of the sampling distribution of the sample mean is obtained by summing the various sample means and dividing the sum by the number of samples. The mean of all the sample means is usually written $\mu_{\bar{X}}$. The μ reminds us that it is a population value because we have considered all possible samples. The subscript $\bar{X}$ indicates that it is the sampling distribution of the sample mean.

$$\begin{aligned}
 \mu_{\bar{X}} &= \frac{\text{Sum of all sample means}}{\text{Total number of samples}} = \frac{\$7.00 + \$7.50 + \dots + \$8.50}{21} \\
 &= \frac{\$162}{21} = \$7.71
 \end{aligned}$$

4. Refer to Chart 8-1, which shows both the population distribution and the distribution of the sample mean. These observations can be made:
- The mean of the distribution of the sample mean (\$7.71) is equal to the mean of the population: $\mu = \mu_{\bar{X}}$.
 - The spread in the distribution of the sample mean is less than the spread in the population values. The sample mean ranges from \$7.00 to \$8.50, while the population values vary from \$7.00 up to \$9.00. Notice, as we increase the size of the sample, the spread of the distribution of the sample mean becomes smaller.
 - The shape of the sampling distribution of the sample mean and the shape of the frequency distribution of the population values are different. The distribution of the sample mean tends to be more bell-shaped and to approximate the normal probability distribution.

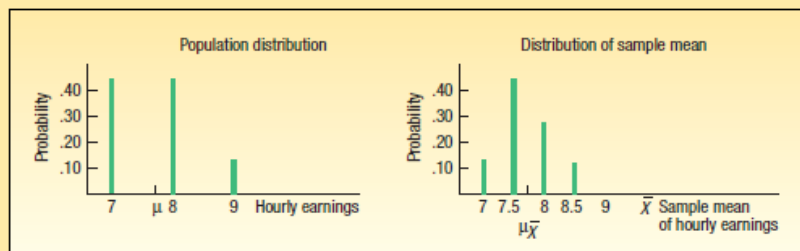


CHART 8-1 Distributions of Population Values and Sample Mean

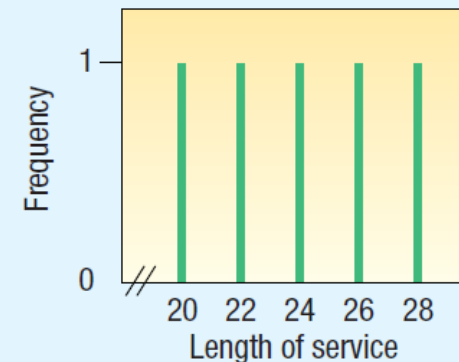
Population mean is equal to the mean of the sample means

Exercise

The lengths of service of all the executives employed by Standard Chemicals are:

Name	Years
Mr. Snow	20
Ms. Tolson	22
Mr. Kraft	26
Ms. Irwin	24
Mr. Jones	28

- Using the combination formula, how many samples of size 2 are possible?
- List all possible samples of 2 executives from the population and compute their means.
- Organize the means into a sampling distribution.
- Compare the population mean and the mean of the sample means.
- Compare the dispersion in the population with that in the distribution of the sample mean.
- A chart portraying the population values follows. Is the distribution of population values normally distributed (bell-shaped)?

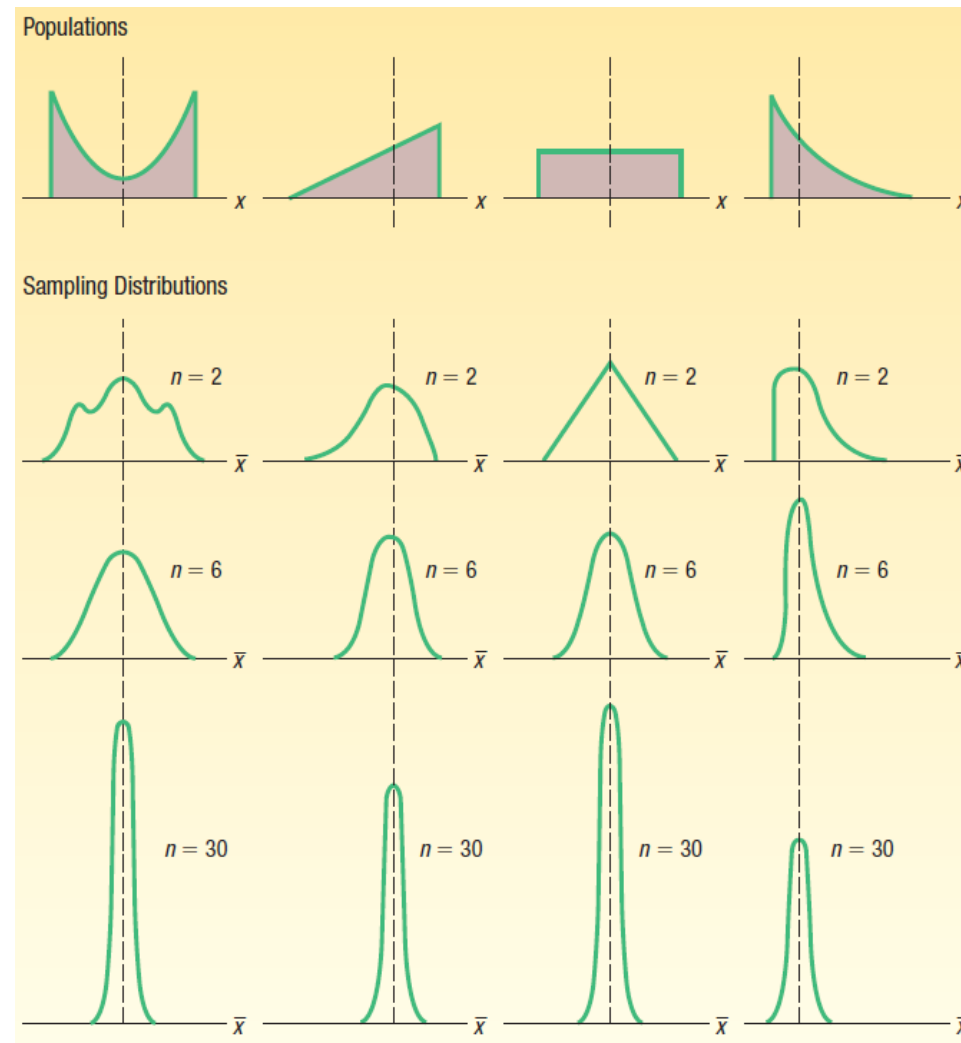


- Is the distribution of the sample mean computed in part (c) starting to show some tendency toward being bell-shaped?

CENTRAL LIMIT THEOREM

It states that If all samples of a particular size are selected from any population, the sampling distribution of the sample mean is approximately a normal distribution. This approximation improves with larger samples.

If the population follows a normal probability distribution, then for any sample size the sampling distribution of the sample mean will also be normal. If the population distribution is symmetrical (but not normal), you will see the normal shape of the distribution of the sample mean emerge with samples as small as 10. On the other hand, if you start with a distribution that is skewed or has thick tails, it may require samples of 30 or more to observe the normality feature. This concept is summarized in Chart



The Standard Normal Probability Distribution

The number of normal distributions is unlimited, each having a different mean (μ), standard deviation (σ), or both. While it is possible to provide probability tables for discrete distributions such as the binomial and the Poisson, providing tables for the infinite number of normal distributions is impossible. Fortunately, one member of the family can be used to determine the probabilities for all normal probability distributions. It is called the **standard normal probability distribution**, and it is unique because it has a mean of 0 and a standard deviation of 1.

Any *normal probability distribution* can be converted into a *standard normal probability distribution* by subtracting the mean from each observation and dividing this difference by the standard deviation. The results are called **z values** or **z scores**.

z VALUE The signed distance between a selected value, designated X , and the mean, μ , divided by the standard deviation, σ .

So, a z value is the distance from the mean, measured in units of the standard deviation.

In terms of a formula:

STANDARD NORMAL VALUE

$$z = \frac{X - \mu}{\sigma}$$

[7-5]

The Standard Normal Probability Distribution

where:

X is the value of any particular observation or measurement.

μ is the mean of the distribution.

σ is the standard deviation of the distribution.

As noted in the preceding definition, a z value expresses the distance or difference between a particular value of X and the arithmetic mean in units of the standard deviation. Once the normally distributed observations are standardized, the z values are normally distributed with a mean of 0 and a standard deviation of 1. So the z distribution has all the characteristics of any normal probability distribution. These characteristics are listed on page 227. The table in Appendix B.1 (also on the inside back cover) lists the probabilities for the standard normal probability distribution. A small portion of this table follows.

The Standard Normal Probability Distribution

To explain, suppose we wish to compute the probability that boxes of Sugar Yummies weigh between 283 and 285.4 grams. we know that the box weight of Sugar Yummies follows the normal distribution with a mean of 283 grams and a standard deviation of 1.6 grams. We want to know the probability or area under the curve between the mean, 283 grams, and 285.4 grams. We can also express this problem using probability notation, similar to the style used in the previous chapter: $P(283 \leq \text{weight} \leq 285.4)$. To find the probability, it is necessary to convert both 283grams and 285.4 grams to z values .

The z value corresponding to 283 is 0, found by $(283 - 283)/1.6$. The z value corresponding to 285.4 is 1.50 found by $(285.4 - 283)/1.6$. Next we go to the table

z	0.00	0.01	0.02	0.03	0.04	0.05	...
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	
.							
.							
.							

The Standard Normal Probability Distribution

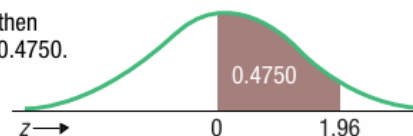
Task (20 Min)

The weekly incomes of shift foremen in the glass industry follow the normal probability distribution with a mean of \$1,000 and a standard deviation of \$100.

1. What is the z value for the income, let's call it X , of a foreman who earns \$1,100 per week? For a foreman who earns \$900 per week?
2. Calculate the likelihood of selecting a foreman and finding his or her income is between \$1,000 and \$1,100?

The Standard Normal Probability Distribution

Example:
If $z = 1.96$, then
 $P(0 \text{ to } z) = 0.4750$.



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990

The Standard Normal Probability Distribution

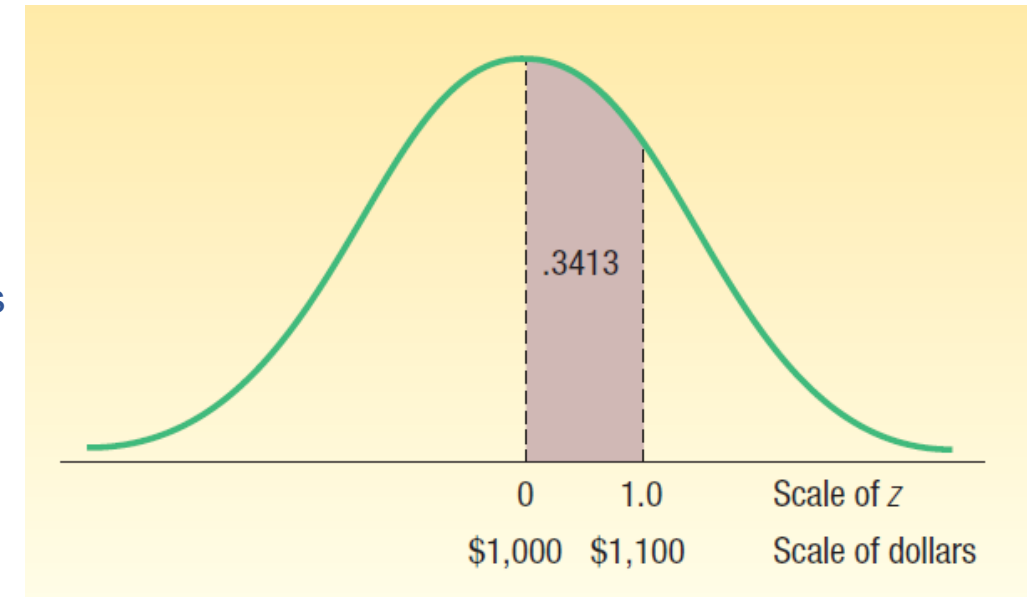
In the example just completed, we are interested in the probability between the mean and a given value. Let's change the question. Instead of wanting to know the probability of selecting at random a foreman who earned between \$1,000 and \$1,100, suppose we wanted the probability of selecting a foreman who earned less than \$1,100. In probability notation, we write this statement as $P(\text{weekly income} \leq \$1,100)$.

The method of solution is the same. We find the probability of selecting a foreman who earns between \$1,000, the mean, and \$1,100. This probability is .3413.

Next, recall that half the area, or probability, is above the mean and half is below.

So the probability of selecting a foreman earning less than \$1,000 is .5000.

Finally, we add the two probabilities, so .3413 + .5000 = .8413. About 84 percent of the foremen in the glass industry earn less than \$1,100 per week. See the following diagram.



The Standard Normal Probability Distribution

Example

Refer to the information regarding the weekly income of shift foremen in the glass industry. The distribution of weekly incomes follows the normal probability distribution, with a mean of \$1,000 and a standard deviation of \$100. What is the probability of selecting a shift foreman in the glass industry whose income is:

1. Between \$790 and \$1,000?
2. Less than \$790?

Solution

We begin by finding the z value corresponding to a weekly income of \$790. From formula (7-5):

$$z = \frac{X - \mu}{s} = \frac{\$790 - \$1,000}{\$100} = -2.10$$

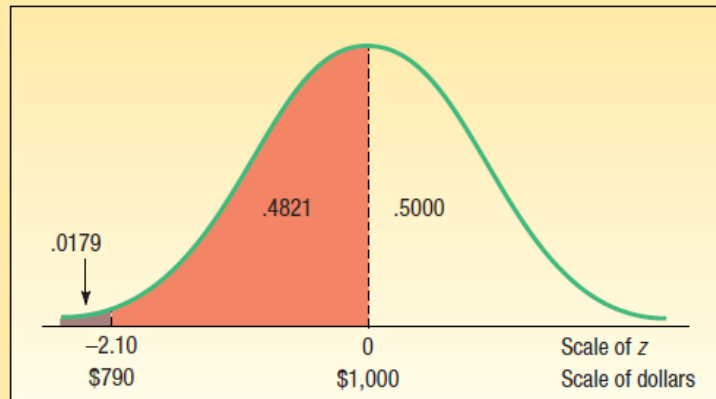
See Appendix B.1. Move down the left margin to the row 2.1 and across that row to the column headed 0.00. The value is .4821. So the area under the standard normal curve corresponding to a z value of 2.10 is .4821. However, because the normal distribution is symmetric, the area between 0 and a negative z value is the same as that between 0 and the corresponding positive z value. The likelihood of finding a foreman earning between \$790 and \$1,000 is .4821. In probability notation, we write $P(\$790 < \text{weekly income} < \$1000) = .4821$.

The Standard Normal Probability Distribution

z	0.00	0.01	0.02
.	.	.	.
.	.	.	.
.	.	.	.
2.0	.4772	.4778	.4783
2.1	.4821	.4826	.4830
2.2	.4861	.4864	.4868
2.3	.4893	.4896	.4898
.	.	.	.
.	.	.	.
.	.	.	.

The mean divides the normal curve into two identical halves. The area under the half to the left of the mean is .5000, and the area to the right is also .5000. Because the area under the curve between \$790 and \$1,000 is .4821, the area below \$790 is .0179, found by $.5000 - .4821$. In probability notation, we write $P(\text{weekly income} < \$790) = .0179$.

This means that 48.21 percent of the foremen have weekly incomes between \$790 and \$1,000. Further, we can anticipate that 1.79 percent earn less than \$790 per week. This information is summarized in the following diagram.



The Standard Normal Probability Distribution

Task (15 Min)

A recent study of the hourly wages of maintenance crew members for major airlines showed that the mean hourly salary was \$20.50, with a standard deviation of \$3.50.

Assume the distribution of hourly wages follows the normal probability distribution. If we select a crew member at random, what is the probability the crew member earns:

- a. Between \$20.50 and \$24.00 per hour?
- b. More than \$24.00 per hour?
- c. Less than \$19.00 per hour?

The Standard Normal Probability Distribution

Example

Recall the distribution of weekly incomes of shift foremen in the glass industry. The weekly incomes follow the normal probability distribution, with a mean of \$1,000 and a standard deviation of \$100. What is the area under this normal curve between \$840 and \$1,200?

Solution

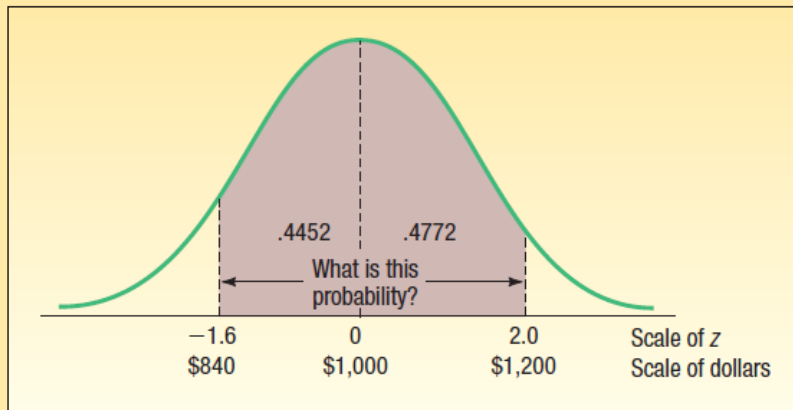
The problem can be divided into two parts. For the area between \$840 and the mean of \$1,000:

$$z = \frac{\$840 - \$1,000}{\$100} = \frac{-\$160}{\$100} = -1.60$$

For the area between the mean of \$1,000 and \$1,200:

$$z = \frac{\$1,200 - \$1,000}{\$100} = \frac{\$200}{\$100} = 2.00$$

The area under the curve for a z of -1.60 is .4452 (from Appendix B.1). The area under the curve for a z of 2.00 is .4772. Adding the two areas: $.4452 + .4772 = .9224$. Thus, the probability of selecting an income between \$840 and \$1,200 is .9224. In probability notation, we write $P(\$840 < \text{weekly income} < \$1,200) = .4452 + .4772 = .9224$. To summarize, 92.24 percent of the foremen have weekly incomes between \$840 and \$1,200. This is shown in a diagram:



The Standard Normal Probability Distribution

Example

Returning to the weekly income distribution of shift foremen in the glass industry ($\mu = \$1,000$, $\sigma = \$100$), what is the area under the normal curve between \$1,150 and \$1,250?

Solution

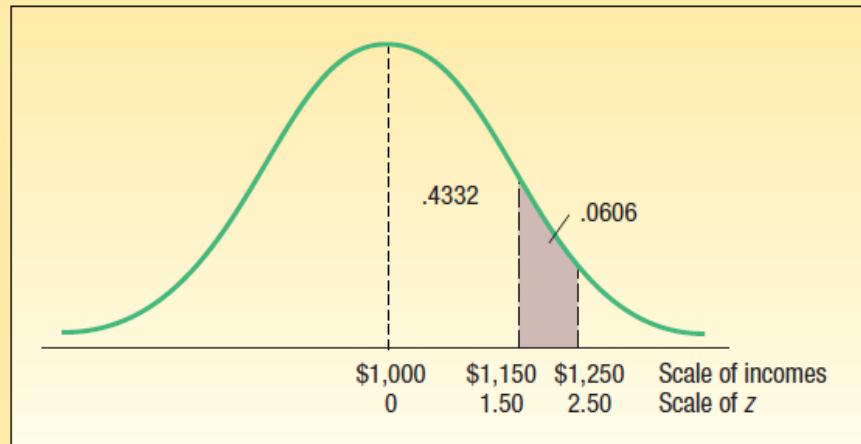
The situation is again separated into two parts, and formula (7-5) is used. First, we find the z value associated with a weekly salary of \$1,250:

$$z = \frac{\$1,250 - \$1,000}{\$100} = 2.50$$

Next we find the z value for a weekly salary of \$1,150:

$$z = \frac{\$1,150 - \$1,000}{\$100} = 1.50$$

From Appendix B.1, the area associated with a z value of 2.50 is .4938. So the probability of a weekly salary between \$1,000 and \$1,250 is .4938. Similarly, the area associated with a z value of 1.50 is .4332, so the probability of a weekly salary between \$1,000 and \$1,150 is .4332. The probability of a weekly salary between \$1,150 and \$1,250 is found by subtracting the area associated with a z value of 1.50 (.4332) from that associated with a z of 2.50 (.4938). Thus, the probability of a weekly salary between \$1,150 and \$1,250 is .0606. In probability notation, we write $P(\$1,150 < \text{weekly income} < \$1,250) = .4938 - .4332 = .0606$.



The Standard Normal Probability Distribution

Task (20 Min)

According to the Internal Revenue Service, the mean tax refund for the year 2007 was \$2,708. Assume the standard deviation is \$650 and that the amounts refunded follow a normal probability distribution.

- a. What percent of the refunds are more than \$3,000?
- b. What percent of the refunds are more than \$3,000 but less than \$3,500?
- c. What percent of the refunds are more than \$2,500 but less than \$3,500?

The number of viewers of *American Idol* has a mean of 29 million with a standard deviation of 5 million. Assume this distribution follows a normal distribution. What is the probability that next week's show will:

- a. Have between 30 and 34 million viewers?
- b. Have at least 23 million viewers?
- c. Exceed 40 million viewers?

The Standard Normal Probability Distribution

Example

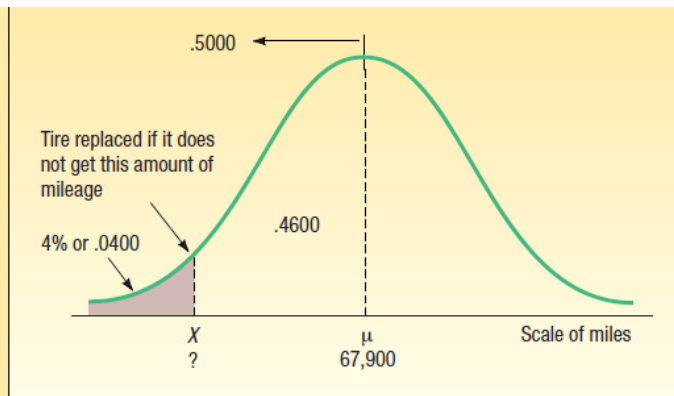


Layton Tire and Rubber Company wishes to set a minimum mileage guarantee on its new MX100 tire. Tests reveal the mean mileage is 67,900 with a standard deviation of 2,050 miles and that the distribution of miles follows the normal probability distribution. Layton wants to set the minimum guaranteed mileage so that no more than 4 percent of the tires will have to be replaced. What minimum guaranteed mileage should Layton announce?

Solution

The facets of this case are shown in the following diagram, where X represents the minimum guaranteed mileage.

The Standard Normal Probability Distribution



Inserting these values in formula (7-5) for z gives:

$$z = \frac{X - \mu}{\sigma} = \frac{X - 67,900}{2,050}$$

Notice that there are two unknowns, z and X . To find X , we first find z , and then solve for X . Notice the area under the normal curve to the left of μ is .5000. The area between μ and X is .4600, found by .5000 - .0400. Now refer to Appendix B.1. Search the body of the table for the area closest to .4600. The closest area is .4599. Move to the margins from this value and read the z value of 1.75. Because the value is to the left of the mean, it is actually -1.75. These steps are illustrated in Table 7-3.

TABLE 7-3 Selected Areas under the Normal Curve

$z \dots$	.03	.04	.05	.06
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.
1.5	.4370	.4382	.4394	.4406
1.6	.4484	.4495	.4505	.4515
1.7	.4582	.4591	.4599	.4608
1.8	.4664	.4671	.4678	.4686

Knowing that the distance between μ and X is -1.75σ or $z = -1.75$, we can now solve for X (the minimum guaranteed mileage):

$$z = \frac{X - 67,900}{2,050}$$

$$z = \frac{X - 67,900}{2,050}$$

$$-1.75 = \frac{X - 67,900}{2,050}$$

$$-1.75(2,050) = X - 67,900$$

$$X = 67,900 - 1.75(2,050) = 64,312$$

So Layton can advertise that it will replace for free any tire that wears out before it reaches 64,312 miles, and the company will know that only 4 percent of the tires will be replaced under this plan.

The Standard Normal Probability Distribution

Hypothesis Testing

- A statistical method to make inferences or draw conclusions about a population based on sample data.
- Elements of a hypothesis test:
 - ❑ **Population:** The entire group being studied.
 - ❑ **Sample:** A subset of the population used to make inferences.
 - ❑ **Null Hypothesis (H_0):** The default assumption (no effect or no difference).
 - ❑ **Alternative Hypothesis (H_1):** The assumption that there is an effect or difference.
 - ❑ **Significance Level (α):** The threshold for rejecting H_0 (commonly 0.05).

Hypothesis Testing

Test Result –	H_0 True	H_0 False
True State H_0 True	Correct Decision	Type I Error
H_0 False	Type II Error	Correct Decision

$$\alpha = P(\text{Type I Error}) \quad \beta = P(\text{Type II Error})$$

Conclusion. If H_0 is rejected, we conclude that H_A is true. If H_0 is not rejected, we conclude that H_0 may be true.



Steps in Hypothesis Testing

1.State the Hypotheses:

1. **Null Hypothesis (H_0)**: This is the hypothesis that there is no effect or no difference. It represents the status quo or a statement of no change.
2. **Alternative Hypothesis (H_1)**: This hypothesis represents what you aim to prove, indicating the presence of an effect or a difference.

2.Choose the Significance Level (α):

- The significance level is the probability of rejecting the null hypothesis when it is true. Common choices are 0.05, 0.01, or 0.10.

3.Select the Appropriate Test:

1. Choose a statistical test based on the data type and the hypotheses. Common tests include:
 1. t-test (for comparing means)
 2. Chi-square test (for categorical data)
 3. ANOVA (for comparing means across multiple groups)
 4. Z-test (for large sample sizes)

4.Collect Data:

- Gather the sample data that will be used for the hypothesis test. Ensure the data is collected in a way that is unbiased and representative of the population.

Steps in Hypothesis Testing

5. Calculate the Test Statistic:

- Using the chosen statistical test, compute the test statistic (e.g., t-value, z-value) based on the sample data.

6. Determine the p-value:

- The p-value indicates the probability of observing the test results under the null hypothesis. It helps to determine the strength of the evidence against the null hypothesis.

7. Make a Decision:

1. Compare the p-value to the significance level (α):
 1. If $p \leq \alpha$: Reject the null hypothesis (H_0).
 2. If $p > \alpha$: Fail to reject the null hypothesis (H_0).

8. Draw Conclusions:

- Interpret the results in the context of the research question. Discuss whether the evidence supports the alternative hypothesis and what it implies for the population.

9. Report the Results:

- Clearly present the findings, including the hypotheses, test statistic, p-value, and any relevant confidence intervals. Provide context and implications of the results.

- Testing hypothesis for the mean μ :

- When the value of sample size (n):

population is normal or not normal
($n \geq 30$)

σ is known

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

σ is not known

$$Z = \frac{\bar{x} - \mu_0}{S / \sqrt{n}}$$

population is normal
($n < 30$)

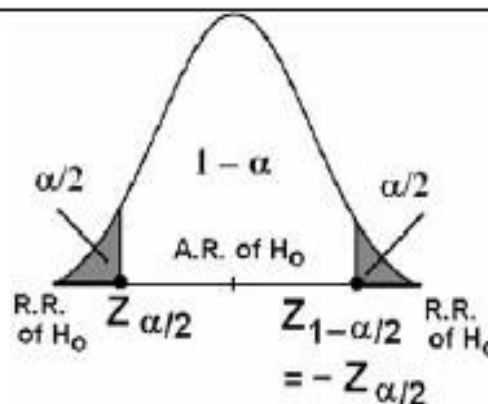
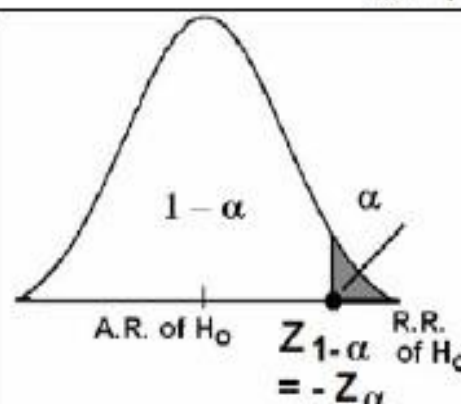
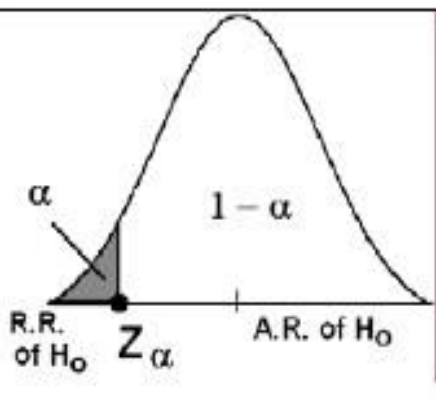
σ is known

$$Z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$$

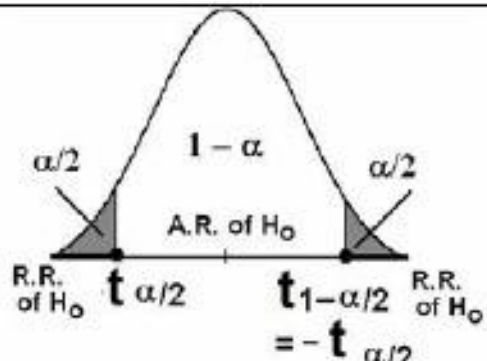
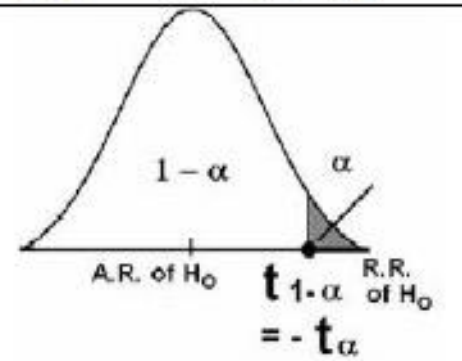
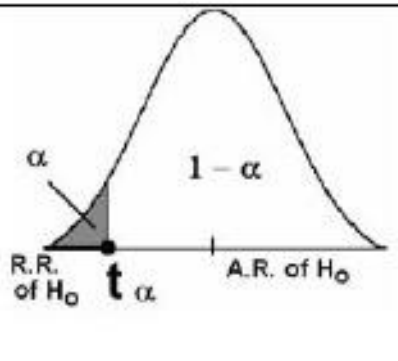
σ is not known

$$T = \frac{\bar{x} - \mu_0}{S / \sqrt{n}}$$

Test Procedures:

Hypotheses	$H_0: \mu = \mu_0$ $H_A: \mu \neq \mu_0$	$H_0: \mu \leq \mu_0$ $H_A: \mu > \mu_0$	$H_0: \mu \geq \mu_0$ $H_A: \mu < \mu_0$
Test Statistic (T.S.)	Calculate the value of: $Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}} \sim N(0,1)$		
R.R. & A.R. of H_0			
Critical value (s)	$Z_{\alpha/2}$ and $-Z_{\alpha/2}$	$Z_{1-\alpha} = -Z_{\alpha}$	Z_{α}
Decision:	We reject H_0 (and accept H_A) at the significance level α if:		
	$Z < Z_{\alpha/2}$ or $Z > Z_{1-\alpha/2} = -Z_{\alpha/2}$ Two-Sided Test	$Z > Z_{1-\alpha} = -Z_{\alpha}$ One-Sided Test	$Z < Z_{\alpha}$ One-Sided Test

Test Procedures:

Hypotheses	$H_0: \mu = \mu_0$ $H_A: \mu \neq \mu_0$	$H_0: \mu \leq \mu_0$ $H_A: \mu > \mu_0$	$H_0: \mu \geq \mu_0$ $H_A: \mu < \mu_0$
Test Statistic (T.S.)	Calculate the value of: $t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t(n-1)$ $(df = v = n-1)$		
R.R. & A.R. of H_0			
Critical value (s)	$t_{\alpha/2}$ and $-t_{\alpha/2}$	$t_{1-\alpha} = -t_{\alpha}$	t_{α}
Decision:	We reject H_0 (and accept H_A) at the significance level α if:		
	$t < t_{\alpha/2}$ or $t > t_{1-\alpha/2} = -t_{\alpha/2}$ Two-Sided Test	$t > t_{1-\alpha} = -t_{\alpha}$ One-Sided Test	$t < t_{\alpha}$ One-Sided Test



cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$	
	one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
	two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df												
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62	
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599	
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924	
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610	
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869	
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959	
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408	
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041	
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781	
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587	
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437	
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318	
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221	
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140	
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073	
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015	
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965	
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922	
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883	
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850	
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819	
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792	
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768	
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745	
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725	
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707	
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690	
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674	
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659	
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646	
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551	
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460	
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416	
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390	
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300	
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291	
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%	
	Confidence Level											

Example 1 - Efficacy Test for New drug

- Drug company has new drug, wishes to compare it with current standard treatment
- Federal regulators tell company that they must demonstrate that new drug is better than current treatment to receive approval
- Firm runs clinical trial where some patients receive new drug, and others receive standard treatment
- Numeric response of therapeutic effect is obtained (higher scores are better).
- Parameter of interest: $\mu_{\text{New}} - \mu_{\text{Std}}$

Example 1 - Efficacy Test for New drug

- **Null hypothesis** - New drug is no better than standard trt

$$H_0 : \mu_{New} - \mu_{Std} \leq 0 \quad (\mu_{New} - \mu_{Std} = 0)$$

- **Alternative hypothesis** - New drug is better than standard trt

$$H_A : \mu_{New} - \mu_{Std} > 0$$

- **Experimental (Sample) data:**

$\bar{y}_{New}$	$\bar{y}_{Std}$
s_{New}	s_{Std}
n_{New}	n_{Std}

Example 1 - Efficacy Test for New drug

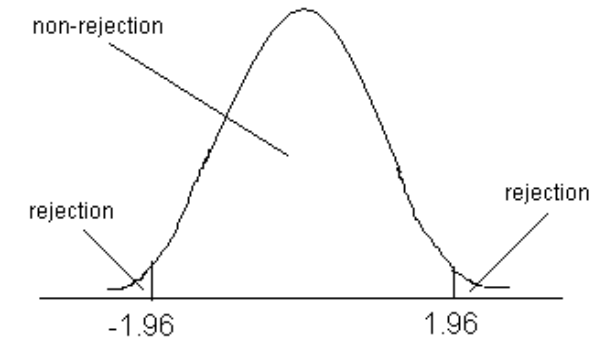
- **Type I error** - Concluding that the new drug is better than the standard (H_A) when in fact it is no better (H_0). Ineffective drug is deemed better.
 - Traditionally $\alpha = P(\text{Type I error}) = 0.05$
- **Type II error** - Failing to conclude that the new drug is better (H_A) when in fact it is. Effective drug is deemed to be no better.
 - Traditionally a clinically important difference (Δ) is assigned and sample sizes chosen so that:
$$\beta = P(\text{Type II error} \mid \mu_1 - \mu_2 = \Delta) \leq .20$$

Example 2 - Mean Age of a Certain Population

- Researchers are interested in the mean age of a certain population.
- A random sample of 10 individuals drawn from the population of interest has a mean of 27.
- Assuming that the population is approximately normally distributed with variance 20, can we conclude that the mean is different from 30 years? ($\alpha=0.05$) .
- If the p - value is 0.0340 how can we use it in making a decision?

Solution

1. **Data:** variable is age, $n=10$, $\bar{x} = 27$, $\sigma^2=20$, $\alpha=0.05$
2. **Assumptions:**
The population is approximately normally distributed with variance 20
3. **Hypotheses:**
 - $H_0 : \mu=30$
 - $H_A : \mu \neq 30$
4. **Test Statistic:** $Z = -2.12$
5. **Decision Rule**
 - The alternative hypothesis is $H_A: \mu \neq 30$
 - Hence we reject H_0 if $Z > Z_{1-0.025} = Z_{0.975}$ or $Z < -Z_{1-0.025} = -Z_{0.975}$
 - $Z_{0.975}=1.96$ (from table D)
6. **Decision:**
 - We reject H_0 , since -2.12 is in the rejection region .
 - We can conclude that μ is not equal to 30
 - Using the p value ,we note that $p\text{-value} = 0.0340 < 0.05$, therefore we reject H_0



CORREL(array1,array2)

Regression

Regression

Regression is a statistical method used to understand the relationship between variables. The primary purpose of regression analysis is to predict a dependent variable (also known as the outcome, response, or target) based on one or more independent variables (also known as predictors, features, or explanatory variables).

Types of Regression:

- **Simple Linear Regression:** One independent variable, straight-line relationship.
- **Multiple Linear Regression:** Multiple independent variables predicting one dependent variable.
- **Logistic Regression:** Used for predicting binary outcomes.

Types of Regression

Linear Regression

- **Linear Regression:** The simplest form of regression, where the relationship between the dependent and independent variable(s) is modeled as a straight line. It can be represented by the equation

$$Y = \beta_0 + \beta_1 x + \epsilon$$

where:

- Y is the dependent variable.
- β_0 is the intercept.
- β_1 is the slope (coefficient of the independent variable x).
- ϵ is the error term.

Predicting House Prices:

Scenario: Predicting the price of a house based on a single feature, such as square footage.

Example: Using historical data on house prices and their square footage to build a model that estimates the price of a new house based on its size.

Multiple Linear Regression

- **Multiple Linear Regression:** An extension of linear regression that uses more than one independent variable. The model can be written as

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \epsilon$$

where:

- Y is the dependent variable.
- β_0 is the intercept.
- β_1 is the slope (coefficient of the independent variable x).
- ϵ is the error term.

Predicting Salary:

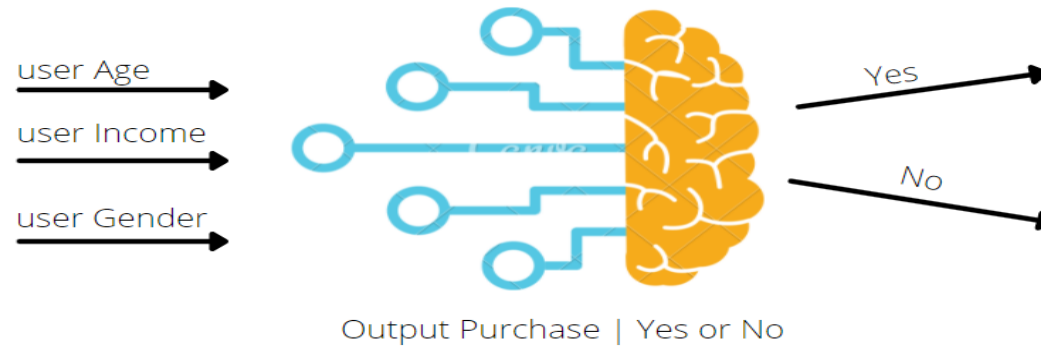
Scenario: Estimating an employee's salary based on multiple factors, such as years of experience, education level, and job role.

Example: Using a dataset that includes years of experience, education level (Bachelor's, Master's, PhD), and job roles (Developer, Manager, Analyst) to predict salaries.

Logistic Regression

Used for binary classification problems, where the dependent variable is categorical (e.g., success/failure, yes/no). It uses the logistic function to model the probability of the default class.

Logistic Regression



Spam Detection:

Scenario: Classifying emails as spam or not spam.

Example: Using logistic regression to build a spam filter that identifies emails as spam based on features like the presence of certain keywords, sender information, and email structure.

Questions?

